Technical Supplement

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

Companion to the inclusive and proton-tagged analysis notes

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Abstract

This supplement contains the final configuration of the CC$\pi^\pm$ NuMI cross-section measurement (Part I), the evidence for its choices (Part II) and the alternatives that were studied and not adopted (Part III).

Charged pions interact hadronically before they stop, so the range-based pion momentum saturates near 0.34 GeV/c and carries little information about the true momentum above about 0.4 GeV/c. Golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, recovery of the daughter energy and the proton-tagged selection were all studied, and none recovers it; a truth-level golden selection performs no better than the classifier, which places the limitation in the physics of pion interactions rather than in the estimator. The pion momentum is therefore measured in three regions. No differential $W_{\pi p}$ result is reported. The topology-level one-bin total cross section of the proton-tagged sample remains measurable, but it is not described as a $W_{\pi p}$ measurement.

The binnings of both families meet a 0.50 migration criterion, and the cross-checks of the extraction (identity closure, the D'Agostini method, ensembles of pseudo-data experiments and the consistency between configurations) are passed. Numbers shared with the analysis notes are checked for consistency automatically (`report/tools/check_tables.py`, `xsec_analyzer/check_staleness.py`). The corrections that shaped the analysis, with the values before and after, are recorded in the change log.

1 Introduction

The inclusive analysis note and the proton-tagged note state what is measured and how. The studies that support their choices are collected here in three parts.

- **Final configuration** (Part I): the status of every result, the final binnings and the rule that selected them, the extraction configuration and uncertainty model, the validation, and the schema of the data release.
- **Evidence for the choices** (Part II): the range estimator of the pion momentum and its saturation; the binnings under the 0.68 and 0.50 migration criteria; the hadronic-mass observables, the proton-tagged selection and its additional smearing; the two-dimensional pairs; the checks of the extraction and the consistency between configurations; the control regions; the selection studies and the automated guards of the pipeline.
- **Rejected alternatives and development history** (Part III): the first binning optimisation, the θ_μ parameterisation, the lower pion threshold, the five-bin p_π response, the details of the golden-pion classifier and of the other pion-momentum estimators, and the cosmic-direction cuts. None of these is used in the submitted results; the change log gives the chronology.

Each section is tagged with the measurement it serves: [INCLUSIVE], [PROTON-TAGGED] or [SHARED]. Terms are defined in the glossary at the end, which is shared with the analysis notes.

Part I
Final configuration

2 Final binnings and the binning rule [SHARED]

Table 1 gives the bin edges of the inclusive observables. The proton-tagged edges are in Table 1 of the proton-tagged note, and those of the two two-dimensional results in their curve files (§5).

Table 1: Bin edges of every observable, from the result files. The binnings meet the 0.50 migration criterion (Sec. 7) and are the same in the three configurations.

Observable	Bins	Edges
p_μ (GeV/c)	6	0.15, 0.3, 0.425, 0.575, 0.825, 1.25, 3*
p_π (GeV/c)	3	0.175, 0.205, 0.26, 1*
$\cos\theta_\mu$	8	-1, 0.4, 0.625, 0.75, 0.825, 0.9, 0.95, 0.975, 1
$\cos\theta_\pi$	5	-1, -0.1, 0.35, 0.6, 0.8, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6

* Numerical plotting boundary only: the upper truth bin is open-ended, and a density drawn over the conventional width represents the bin integral.

Each one-dimensional binning is selected by the following rule, established by the binning review (§7):

1. Generate candidate partitions: for every number of bins up to twelve, the maximin partition, which has the largest worst-bin migration diagonal over FHC and RHC, found by dynamic programming on a fine grid (`report/tools/maximin_binning.py`).
2. Require the event floor: at least 70 selected FHC signal events per bin at data exposure in the inclusive family, and 50 in the proton-tagged family.

3. Require every diagonal of the column-normalised migration matrix to be at least 0.50, in FHC and in RHC, on the analysis sample (§6.1); the combined configuration inherits the binning.
4. Extract the candidates that pass in full, in the three configurations: universes, covariance, unfolding and closure.
5. Compute the Wiener filter factors of each extraction and the trace of A_C, their sum, which is the number of degrees of freedom that the data determine.
6. Select the fewest bins that reach 90% of the largest trace, averaged over FHC and RHC; ties go to the larger worst diagonal.
7. Check the closure and the per-bin uncertainties of the selected binning.
8. Freeze the binning.

Four adopted binnings differ from the one the rule selects, for the reasons given in §7: $\cos\theta_\pi$ and $\theta_{\mu\pi}$ keep five bins, $\delta\alpha_T$ keeps its split at 120°, and the proton-tagged p_π keeps two regions. The 0.68 criterion of the BNB CC1π measurement is the precedent for the migration criterion and provides a stricter comparison point; §7 gives the consequences of both. Everything in Part III, including the first binning optimisation, the five-bin p_π response and the θ_μ parameterisation, is historical and is not used in the submitted results.

A result can close while it is determined largely by the modelled migrations or by the regularisation, so the migration criterion and the Wiener filter factors are applied as separate criteria of the information a binning retains.

3 Extraction configuration and uncertainty model [SHARED]

Unfolding. Every differential result is unfolded with the Wiener-SVD method. The regularisation matrix is the second derivative of the cross-section density, built with the actual bin widths: on binnings as non-uniform as these, the uniform stencil reduces A_C to a matrix that passes the normalisation and discards the shape (§11). The additional-smearing matrix A_C of every result is released with it, and every prediction is smeared by it before it is compared with the result. The one-bin total cross sections are extracted with the Wiener filter off, $A_C \equiv 1$ (§14). The D'Agostini iterative method is a cross-check (§12).

Normalisation. Each run is normalised to its recorded exposure: the simulation of every run is scaled by the ratio of its beam-on to its simulated POT, and the beam-off samples by the per-run gate ratios (Sec. 5 of the analysis note). The flux is the summed $\nu_\mu + \bar{\nu}_\mu$ flux integrated above 60 MeV: FHC 6.81159, RHC 6.44646 and combined configuration 6.59691×10^{-10} cm^{-2} POT^{-1}, each configuration normalised with its own value. The 60 MeV convention includes the flux below the pion-production threshold, so it lowers the flux-averaged total relative to a threshold matched to the spectrum (Sec. 4.4 of the analysis note). The target count is $N_{\mathrm{Ar}} = 8.710 \times 10^{29}$ argon nuclei in the fiducial volume.

Covariance. The official covariance of every result, `cov_total_plusMCS.txt`, is the framework covariance, `cov_total.txt`, plus the MCS muon-momentum-scale term. The framework covariance sums the flux (PPFX universes), cross-section (GENIE multisim and unisims), hadron re-interaction, detector-variation, POT and target-count terms and the statistical terms of the simulation, the beam-off samples and the data, each propagated through the unfolding and released separately (§5). The detector-variation term applies the fractional change of each variation relative to the detector-variation CV sample to the nominal prediction (Sec. 12 of the analysis note). The MCS term is the half-difference, taken as fully correlated, of the results

obtained when the MCS-measured muon momenta of the pseudo-data are scaled by ±5% on the data side and unfolded with the nominal response. It is computed outside the framework, is zero for the results that do not use the reconstructed muon momentum, and is included in every quoted uncertainty and every summary table, among them the systematic breakdowns of Tables 20–22 (Sec. 12.3 of the analysis note). The closure χ^2 and the biases of the robustness tests are quoted with the framework covariance, since they test the extraction rather than the uncertainty of the data. Three terms lie outside the official covariance: the beamline-geometry flux term, the uncertainty of the per-run beam-off trigger ratios and that of the use of the Run-1 beam-off sample for Run 2.

4 Validation [SHARED]

The tests of the extraction go from the implementation to the model. The identity closure and the pseudo-data closure test the implementation (§13). Every extraction closes: the inclusive ones with $\chi^2 \leq 3.12/8$ and integral ratios of 0.959–1.035, the proton-tagged ones with $\chi^2 \leq 7.28/4$ (RHC $\cos\theta_\pi$) and ratios of 0.91–1.21, and the two two-dimensional results with cell ratios of 0.62–1.33. A result can close while it is determined largely by the modelled migrations or by the regularisation, so the migration criterion and the trace of A_C are applied as separate criteria (§2).

Ensembles of 100 pseudo-experiments test the size of the statistical uncertainties (§17). Seven inclusive and ten proton-tagged extractions were run, the proton angles in every configuration and the two two-dimensional results among them; the pull widths are 0.92–1.04 and 0.92–1.09, and the 68% coverages 64–72% and 64–73%.

Pseudo-data from two reweighted interaction models, GENIE multisim universe 545 and the $\Delta \to N\pi$ angular unisim, test the robustness to a model that reweighting can reach (Sec. 13.3 of the analysis note, Sec. 5.3 of the proton-tagged note). The largest bias is 1.21σ, in one sparse inclusive bin (FHC $\cos\theta_\mu$); in the proton-tagged sample the low δp_T and p_n bins move by -1.08σ and -1.04σ. The proton angles and the two two-dimensional results stay within 0.88σ.

No closure test with an independent generator has been made; the NuWro overlay available so far disagrees with standalone NuWro and is not used (Sec. 15 of the analysis note). The control regions, which test the background model, have been studied with pseudo-data only (§19).

5 Data-release schema [SHARED]

The numerical results are in `report/data_release/` (Table 2). Each extraction is identified by a tag `<family>_<CFG>_<observable>`: the family is `incl` or `1p`, the configuration `FHC5`, `RHCFULL` or `COMB`, and the observable is the key used in `index_extractions.tsv`, for example `incl_FHC5_pmu` or `1p_COMB_thetap`. The two two-dimensional results have the tags `1p_COMB_thetap_dpt2d` and `incl_COMB_costhpi_costhmu2d`. Cross sections are in 10^{-38} cm^2/Ar per unit of the variable, or per unit of each of the two variables for the two-dimensional results. Uncertainties and correlations are taken from the covariance files.

Table 2: Files of the data release in `report/data_release/`; `<tag>` is the tag of an extraction.

File	Content
`curves_<tag>.tsv`	One row per bin: `bin`, `low`, `high`, `width`, `unfolded_data`, `stat_err`, then the truth, tune and generator predictions, all already smeared by A_C. The values are densities; multiply by `width` to integrate. The two-dimensional files carry `inner_low`, `inner_high`, `outer_low`, `outer_high` and `area` in place of `low`, `high` and `width`, and order the analysis bins outer slice by outer slice, with the inner variable fastest.
`A_C_<tag>.tsv`	The additional-smearing matrix A_C: row i is the smeared bin, column j the true bin. A prediction p_j, the cross section integrated over true bin j, is compared with the result as smeared$_i = \sum_j A_C[i][j]\, p_j/\text{width}_i$, with the area in place of the width for the two-dimensional results.
`cov/<tag>/cov_total_plusMCS.txt`	The official covariance: the framework covariance plus the MCS momentum-scale term.
`cov/<tag>/cov_total.txt`	The framework covariance.
`cov/<tag>/cov_<source>.txt`	The components by source: flux, detector variations, cross section, re-interaction, POT, target count, simulation, beam-off and data statistics, MCS momentum scale.
`cov/<tag>/unfolded_signal.txt`	The unfolded bin integrals.
`total_xsec.tsv`	The one-bin total cross sections, with their uncertainty breakdown.
`index_extractions.tsv`	Every extraction, with its status.
`../status.tsv`	The status of every product (in `report/`, next to the release).

Part II
Evidence for the choices

6 Pion momentum: the range estimator and its saturation [INCLUSIVE]

The inclusive p_π is measured in three regions with edges $[0.175, 0.205, 0.26, \infty)$ GeV/c, and the proton-tagged p_π in two regions, $[0.175, 0.205, \infty)$ GeV/c. The five-bin configurations are diagnostics, not results. The golden-pion, MCS, calorimetric, regression and daughter-recovery studies of this section support the coarse range-based measurement.

6.1 Track sample and analysis sample

Migration diagonals of the pion momentum are quoted on two samples, which are not interchangeable. The track sample is every truth-matched pion track passing loose quality requirements (purity > 0.5, track score > 0.5, length > 5 cm), from events of any topology, 126 518 tracks in total; the golden-pion studies use it, and the estimator studies use truth-matched pion tracks of the same kind. The analysis sample is the events passing the full CC1μ1$\pi^\pm$ selection and satisfying the signal definition, about 10 700 per beam mode; the adopted binnings, the binning studies and the hadronic-mass study use it. The analysis sample has the better response, because the selection removes the events whose pion is poorly reconstructed.

6.2 Pion-momentum bias and the conditioning of the p_π response

The p_π response is the least diagonal of any observable in the analysis. The cause is physical, and it limits what the measurement can say about the pion momentum spectrum.

6.2.1 Mass hypothesis and hadronic truncation.

The reconstructed pion momentum stored in the ntuple, `candidate_pion_mom_reco`, is `trk_range_muon_mom_v`: a range-based momentum evaluated under the muon mass hypothesis and applied to a pion track. Two distinct effects follow, and only the first is corrected.

First, the mass hypothesis biases the estimate low, because for a given range a pion carries more momentum than a muon. The size is $\sim$17–19% from the CSDA range scaling, largely independent of momentum. This effect is corrected in the reco bin definitions rather than in the branch: `ccpi_ppi_bin_config_opt.txt` applies a kinetic-energy-preserving $\mu \to \pi$ mass swap, $E_\mu \to T \to E_\pi \to p_\pi$, before binning. In the lowest true bin $\langle p^{\mathrm{reco}}/p^{\mathrm{true}} \rangle$ goes from 0.888 with the raw branch to 0.976 with the correction.

Second, charged pions interact hadronically before coming to rest, and nothing corrects for this. The reconstructed track then ends at the interaction point rather than the stopping point, so the visible range, and any momentum inferred from it, is truncated. The effect grows rapidly with momentum and dominates at high momentum.

6.2.2 Saturation of the reconstructed momentum.

Table 3 and Figure 1 show the result: as the true momentum rises by a factor of four and a half, the reconstructed momentum barely moves, saturating at ≈ 0.34 GeV/c. Above that the estimator carries almost no information about the energy of the pion.

Table 3: Pion momentum reconstruction versus truth, selected signal, Run-1 FHC. "raw" is the `candidate_pion_mom_reco` branch; "corr." also applies the $\mu \to \pi$ mass swap used by the reco bin definitions. The mass correction removes the bias at threshold; nothing removes the saturation above it.

true p_π [GeV/c]	N	$\langle p^{\mathrm{true}} \rangle$	$\langle p^{\mathrm{corr}} \rangle$	raw/true	corr./true
0.175–0.250	528	0.210	0.205	0.888	0.976
0.250–0.300	270	0.276	0.238	0.790	0.863
0.300–0.350	254	0.325	0.253	0.716	0.779
0.350–0.400	265	0.374	0.270	0.664	0.721
0.400–0.500	403	0.443	0.279	0.581	0.629
0.500–0.600	252	0.548	0.307	0.520	0.560
0.600–0.800	272	0.688	0.326	0.441	0.473
0.800–1.200	164	0.951	0.336	0.329	0.353

6.2.3 Effect on the five-bin response matrix.

On the analysis sample of Table 3, migrating the five-bin scheme with the corrected estimator gives a nearly lower-triangular matrix: the diagonal fraction falls from 94.3% in the first true bin to 36.5, 23.1, 10.4 and finally 6.2% in the last. For the highest true bin ($p_\pi > 0.55$ GeV/c) only 6.2% of events are reconstructed in the corresponding reco bin while 32.6% fall into the lowest one. Every true bin above the first therefore deposits most of its events into reco bin 1, the "pile-up into the first bin" of Fig. 11.

6.2.4 Treatment in the BNB CC1$\pi^\pm$ analysis.

The published BNB CC1$\pi^\pm$Xp analysis (internal note v0.91) uses an identical estimator: it too takes "the momentum of pion candidates from range". Its treatment is summarised here and was followed by this analysis.

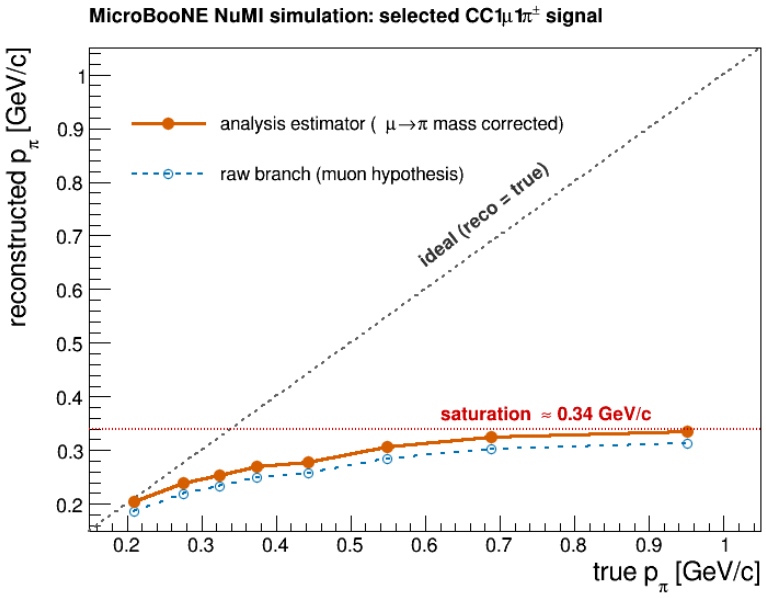


Figure 1: Mean reconstructed against mean true pion momentum, selected signal. The analysis estimator (filled, $\mu \to \pi$ mass corrected) tracks truth at threshold but flattens out near 0.34 GeV/c: a true 0.95 GeV/c pion reconstructs at the same place as a true 0.55 GeV/c one. The raw `candidate_pion_mom_reco` branch (open) sits systematically below it by the uncorrected mass-hypothesis offset. The saturation is caused by hadronic interaction truncating the track range, and no range-based estimator can recover it.

That analysis does three things this one does not. First, it defines a *golden pion*: one that comes to rest and leaves a Bragg peak, as opposed to one that scattered or interacted. Second, it trains a dedicated golden-pion BDT and applies it as an extra cut for the pion-momentum differential cross section only, leaving the total and angular results on the generic selection. The BDT separates golden from non-golden pions using track "wiggliness" (the standard deviation of the angular difference between directions of adjacent points along the fitted track, a direct handle on scattering), the number of reconstructed descendant particles, the track–shower score, and the Bragg likelihood ratio. The note quotes the cost and benefit: efficiency falls from 20.3% to 9.5% and purity is largely unchanged (51.9 → 49.5%), while the golden fraction of the selected signal rises from 36.3% to 67.7%. The same logic is applied to the muon momentum, restricting that cross section to contained muons (8.4% efficiency, 56.1% purity) so that it uses range only.

Third, and most directly comparable, the binning is chosen against an explicit criterion: each bin must hold at least 100 predicted MC events and the diagonal element of the migration (smearing) matrix must exceed 0.68. The resulting pion-momentum edges are $0.10, 0.16, 0.19, 0.22, 0.60$ GeV/c with an overflow bin: fine structure only where the estimator still tracks truth, then a single wide bin from 0.22 to 0.60 and everything above 0.60 in overflow.

The five-bin scheme fails that criterion: its diagonal fractions are $94.3, 36.5, 23.1, 10.4$ and 6.2%, so one of five bins passes 0.68 and the top three fail it by a wide margin. Its edges (0.175–0.550 in five bins, open above) place four boundaries inside the region where the reconstructed momentum has saturated.

6.2.5 Consequences for the unfolding.

The saturation does not bias the extracted cross section, because the response matrix contains it and the unfolding inverts it, but it makes the p_π result depend more on the regularisation than any other observable. With 6% diagonal strength in the top bin, the result above $\sim$0.5 GeV/c is

effectively inferred from the response model rather than measured.

Adopted binning: three p_π regions

The adopted binning has three regions, $[0.175, 0.205]$ and $[0.205, 0.26]$ GeV/c and everything above. Merging raises the migration diagonals from 96/35/21/10/6% (full analysis sample; 94.3/36.5/23.1/10.4/6.2% on Run 1 alone, Table 3) to 91/51/52% in FHC and 90/53/53% in RHC, which meets the 0.50 criterion of the analysis in every region (§7). Under the 0.68 criterion of the BNB measurements only two regions pass, $[0.175, 0.205]$ GeV/c and everything above, at 91/76%. These diagonals are measured on the analysis sample (§6.1); the golden-pion study of Sec. 6.2.6 uses the looser track sample, whose diagonals are lower, and the two sets are not interchangeable.

The first boundary follows from a scan of the two-region binning on the analysis sample. The worst diagonal is always that of the upper region, and it falls monotonically as the boundary rises, so the binding constraint is the width of the first interval rather than the diagonal itself:

boundary	first-bin width	diagonal, bin 1	diagonal, bin 2
0.185	10 MeV/c	79.7%	85.4%
0.195	20 MeV/c	88.8%	80.0%
0.200	25 MeV/c	90.4%	77.7%
0.205	**30 MeV/c**	**91.0%**	**75.5%**
0.220	45 MeV/c	93.3%	69.1%
0.250	75 MeV/c	95.6%	56.1%
0.300	125 MeV/c	96.3%	39.0%

Both regions meet 0.68 anywhere between roughly 0.19 and 0.22, so the choice within that window is set by width rather than by the criterion. 0.205 is the narrowest first interval with precedent: 30 MeV/c matches the narrowest bin used in the BNB CC1π analysis. Going narrower trades first-bin diagonal for upper-bin diagonal, and at 10 MeV/c the first bin itself falls to 80% on only 325 events. An exhaustive scan over all edge sets (`macros/ppi_binning_scan.C`) returns 0.205 as the best first boundary for every number of bins tried, and in the proton-tagged sample as well.

The second boundary costs 25 points on the worst diagonal. The best three-region binning, with edges 0.205 and 0.26 GeV/c from the maximin scan of §7, reaches 91/51/52%, against 91/76% for two regions: it meets 0.50 but not 0.68. Two 30 MeV/c intervals followed by an open region (0.205, 0.235) are worse at 91/39/62%, and degrade the upper region too, from 76% to 62%: narrowing its reconstructed range pushes events of high true momentum out of their own bin. Four regions reach only 91/39/39/42%. Above $\sim$0.25 GeV/c the reconstructed momentum barely varies, so a further boundary there separates almost nothing. The same ordering holds under a golden-pion cut (67/52/46%), so it is not an artefact of the selection.

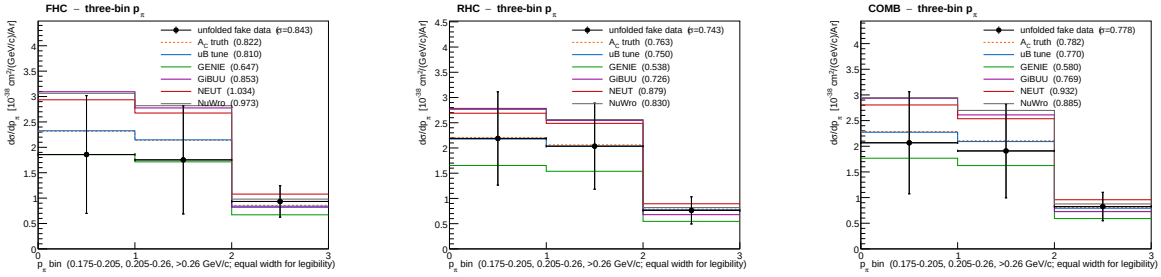
6.2.6 Closure.

The three-region scheme is carried through the full chain (universe construction, Wiener–SVD unfolding and systematics) and closes in all three configurations (Table 4). A five-bin extraction on the same sample also closes, at $\chi^2 \approx 2/5$: closure cannot expose a response that carries no shape information, whereas the migration criterion can.

All three configurations close, with χ^2 of 0.45, 0.01 and 0.13 on three bins and unfolded-to-truth integral ratios of 1.026, 0.973 and 0.995 in FHC, RHC and combined. GENIE is the lowest prediction in every region of every configuration, as in the other observables. The released A_C for FHC is $\left(\begin{smallmatrix} 0.30 & 0.21 & 0.01 \\ 0.41 & 0.41 & 0.02 \\ -0.49 & 0.04 & 0.87 \end{smallmatrix}\right)$ (`A_C_incl_FHC5_ppi3bin.tsv`): the two near-threshold regions are mixed strongly, and together they carry one component that the data determine, so the trace is 1.58 and two filter factors exceed 0.5. The first two regions are therefore measured as a pair rather

Table 4: Three-region p_π result for all three beam configurations, $d\sigma/dp_\pi$ in 10^{-38} cm^2/(GeV/c)/Ar. Bin 1 is $[0.175, 0.205]$ GeV/c, bin 2 $[0.205, 0.26]$ GeV/c and bin 3 everything above. All curves are A_C-smeared, so they live in the same space as the measurement. Generated from the release (`report/tools/ppi2bin_tables.py`: closure sidecars and `curves_incl_*_ppi3bin.tsv`); the uncertainty row is the total.

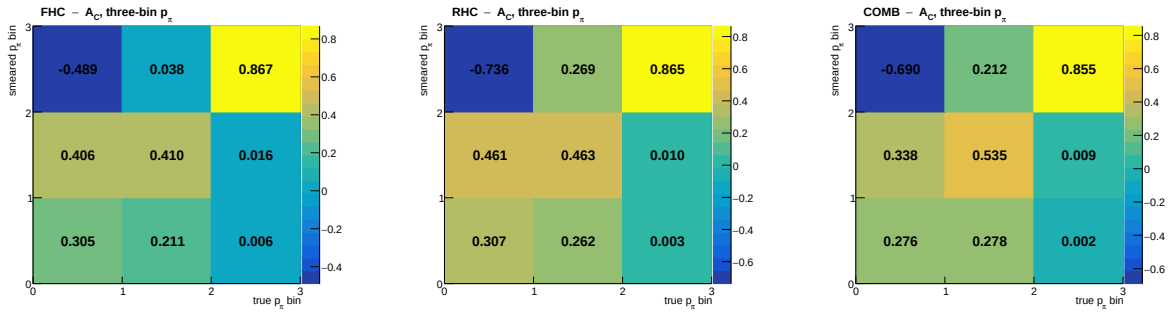
	FHC				RHC				Combined			
	bin 1	bin 2	bin 3	σ_{int}	bin 1	bin 2	bin 3	σ_{int}	bin 1	bin 2	bin 3	σ_{int}
unfolded data	1.86	1.75	0.93	0.843	2.19	2.04	0.76	0.743	2.07	1.91	0.83	0.778
uncertainty	$\pm$1.16	$\pm$1.07	$\pm$0.31		$\pm$0.93	$\pm$0.85	$\pm$0.27		$\pm$0.99	$\pm$0.91	$\pm$0.28	
A_C truth	2.32	2.14	0.86		2.21	2.06	0.79		2.29	2.11	0.81	
uB tune	2.33	2.15	0.84		2.18	2.03	0.77		2.27	2.09	0.79	
GENIE	1.85	1.71	0.67		1.65	1.54	0.55		1.77	1.62	0.59	
GiBUU	3.09	2.78	0.82		2.78	2.55	0.68		2.93	2.61	0.73	
NEUT	2.94	2.68	1.08		2.69	2.49	0.89		2.80	2.54	0.96	
NuWro	3.06	2.82	0.98		2.76	2.56	0.82		2.95	2.70	0.88	
χ^2/ndf	0.45/3 ($p = 0.93$)				0.01/3 ($p = 1.00$)				0.13/3 ($p = 0.99$)			



(a) FHC (b) RHC (c) Combined

Figure 2: The three-region p_π cross section, the graphical form of Tab. 4. Unfolded pseudo-data (points, with the full systematic uncertainty) against the A_C-smeared truth, the uB tune and the four generators; every curve is A_C-smeared, so all of them live in the same space as the measurement. The regions are drawn with equal width for legibility: they are 0.030, 0.055 and 0.74 GeV/c wide, so area on these panels is not proportional to cross section, and the integrated σ is printed on each panel instead.

than separately, and a prediction is compared through the full A_C. The RHC and combined matrices mix the two regions even more (0.31/0.26 and 0.46/0.46 in RHC); the upper-region diagonal is 0.86–0.87 in all three. The total fractional uncertainty is 52% in FHC, the largest of the inclusive observables, because the regions each carry a large share of the flux normalisation with little statistical averaging. Figure 3 shows A_C for all three configurations.



(a) FHC (b) RHC (c) Combined

Figure 3: The additional-smearing matrix A_C for the three-region p_π scheme, in the convention $\hat{x}_i = \sum_j (A_C)_{ij}\, t_j$ so that a column is a true bin and a row is the smeared bin it feeds. A_C is not normalisation-preserving, so its rows do not sum to unity. The negative entries of smeared bin 3, drawn at the top, are the regularisation borrowing between the near-threshold regions and the upper one; they are why the truth curves in Fig. 2 must be A_C-smeared before being compared with the measurement rather than plotted raw.

At 91/51/52% the three regions meet the 0.50 criterion. The result remains a coarse measurement: the near-threshold regions are measured as a pair, and above $\sim$0.34 GeV/c the reconstructed momentum saturates, so the upper region is an integral over a wide range rather than a differential point. The scheme provides a result with a defensible response rather than a competitive shape measurement.

Golden-pion classifier

Following the BNB recipe, a golden-pion BDT was trained and evaluated with the macros `golden_pion_train.C` and `golden_pion_compare.C` of `macros/`. The truth label is `mc_end_p` < 0.01 GeV/c (the pion ranged out), joined to reconstruction by matching `backtracked_p` to `mc_p`, which pairs 100% of candidate tracks with no ambiguity. The inputs are the three deflection variables, the four Bragg likelihood ratios, the track–shower score and the descendant count. Training used 126 518 labelled tracks of the track sample (§6.1) from Runs 1, 2 and 4; the split is by run, and all numbers below are on Run 2, which the classifier did not see in training.

6.2.7 Classifier performance.

The ROC integral is 0.758. At 44.8% efficiency the classifier raises the golden fraction from 43.1% to 64.9%, against the BNB 36.3 $\rightarrow$ 67.7% at roughly 47% efficiency, so it reproduces the BNB benchmark.

6.2.8 Effect on the p_π migration.

Table 5 gives the migration diagonal in the five bins as the cut is tightened. The first bin passes and continues to pass; no other bin reaches 0.68 at any working point, even at 15% efficiency.

Table 5: Golden-pion BDT: efficiency, achieved golden fraction, and the p_π migration diagonal per true bin, on the held-out run of the track sample (§6.1). The BNB criterion for a usable bin is a diagonal above 0.68.

BDT cut	eff.	golden frac.	b1	b2	b3	b4	b5
none	100%	43.1%	99.2	23.7	13.2	6.4	5.5
> -0.05	57.0%	58.1%	99.1	29.4	19.7	11.2	7.1
> 0.00	44.8%	64.9%	99.1	33.1	24.5	15.4	9.1
$> +0.10$	24.6%	79.4%	99.1	42.9	36.8	29.3	15.7
$> +0.15$	15.2%	84.5%	99.2	46.9	42.2	36.1	19.5
perfect (truth) golden selection			98.9	36.1	24.1	14.2	8.9

6.2.9 Truth-level ceiling.

The last row of the table is to be compared with the row of matching efficiency rather than with the tightest cut. Perfect truth-level golden selection keeps 43.1% of tracks and gives $36.1, 24.1, 14.2, 8.9\%$ in bins 2–5; the classifier at the matching 44.8% efficiency gives $33.1, 24.5, 15.4, 9.1\%$. They agree bin by bin. The classifier therefore already performs as well as perfect knowledge of which pions stopped, and a better golden-pion classifier cannot improve the response further. The two tightest working points exceed the truth-level row only because a hard cut on the classifier preferentially keeps long tracks, and so cuts on the reconstructed momentum itself; they are not used (§27).

6.2.10 Summary of the classifier study.

The classifier matches the published benchmark, but the p_π differential measurement above $\sim$0.23 GeV/c cannot be made to meet the BNB binning criterion with a range-based estimator, with or without a golden-pion cut. The BNB analysis meets that criterion because its resolved bins (0.10–0.16, 0.16–0.19, 0.19–0.22) all lie below the 0.175 GeV/c threshold of this analysis, in the region where range still tracks momentum; above 0.22 it uses a single wide bin and an overflow. The options for this analysis were therefore to lower the pion threshold toward 0.10 GeV/c and resolve the region where the estimator works (tested and not adopted, §25), to report a coarse p_π result with an open upper region (adopted, §6.2.5), or to replace the range-based estimator (§6.3). The golden-pion BDT can contribute to any of these but does not by itself solve the problem.

The limitation described in this subsection concerns attempts to recover a finer multi-bin p_π spectrum, and its numbers are for the loose truth-matched track sample. It does not apply to the adopted three-region result on the analysis sample (§6.2.5, Table 4), whose migration diagonals are 91%, 51% and 52%.

6.2.11 Scope of the p_π result.

Two routes to a finer p_π measurement were tested. The golden-pion classifier does not repair the response even at perfect truth-level selection (above), and the proton tag does not change the response appreciably (below). With merged regions the inclusive selection satisfies the binning criterion, so p_π is retained in the inclusive measurement; what it cannot support is a five-bin shape measurement. The other inclusive observables are unaffected: the angular variables and p_μ retain their shape information, and the total cross section does not use this estimator.

6.2.12 Proton-tagged response.

The second route to a sharper p_π response is the proton tag. It was measured for FHC, RHC and the combined configuration: at the same two-bin edges the proton-tagged response

is indistinguishable from the inclusive one, and marginally worse rather than better. The migration-diagonal test gives

selection	config.	N_{sig} bin 1	bin 1 $[0.175, 0.205]$	bin 2 (> 0.205)	both > 0.68?
inclusive `CC1mu1piXp`	FHC	1056	91.0%	75.5%	yes
proton-tagged	FHC	469	**88.9**%	**74.2**%	**yes**
`CC1mu1pi1p`	RHC	621	**89.4**%	**75.1**%	**yes**
	comb.	1090	**89.2**%	**74.7**%	**yes**

measured on the `numuMC` overlays named in the corresponding `file_properties` lists, selected signal (analysis sample), with the reconstructed momentum mass-corrected as in the bin configuration. All four rows clear the criterion in both bins, and the proton-tagged values lie about two points below the inclusive ones. The three proton-tagged configurations agree to better than a percentage point, so the response is a property of the selection rather than of a particular exposure, but the proton tag does not improve it.

This is expected. The pion-momentum response is set by whether the pion ranges out or interacts hadronically before stopping, which is a property of the pion's own passage through the argon. Requiring a reconstructed proton at the vertex changes the event sample, the purity and the efficiency, but not what happens to the pion downstream, so there is no mechanism by which it would sharpen the momentum response, and the measurement confirms this.

The proton-tagged p_π extraction is therefore the pion-momentum cross section of an exclusive final state that also yields W and TKI, a Validation-only product, not a repair of the inclusive one. The inclusive p_π in three regions also meets the adopted criterion.

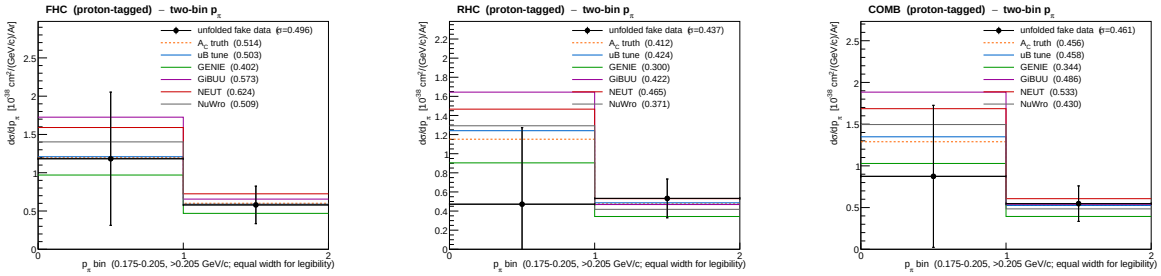
Table 6: Two-bin p_π on the proton-tagged `CC1mu1pi1p` selection, all three beam configurations. $d\sigma/dp_\pi$ in 10^{-38} cm^2/(GeV/c)/Ar; bin 1 is $[0.175, 0.205]$ GeV/c, bin 2 everything above. All curves are A_C-smeared. The proton-tagged `xsec` configs declare only the tune and the pseudo-data, so no generator rows appear. Regenerated from the release (`report/tools/ppi2bin_tables.py`); the uncertainty row is the total.

	FHC		RHC		Combined	
	bin 1	bin 2	bin 1	bin 2	bin 1	bin 2
unfolded data	1.183	0.580	0.472	0.532	0.874	0.547
uncertainty	$\pm$0.870	$\pm$0.245	$\pm$0.801	$\pm$0.203	$\pm$0.851	$\pm$0.213
A_C truth	1.201	0.601	1.152	0.475	1.289	0.525
uB tune	1.209	0.587	1.241	0.486	1.348	0.526
unfolded/truth	0.98	0.96	0.41	1.12	0.68	1.04
σ_{int}	0.496		0.437		0.461	
χ^2/ndf	0.01/2 ($p = 1.00$)		1.23/2 ($p = 0.54$)		0.34/2 ($p = 0.84$)	

All three extractions close, with χ^2 of 0.01, 1.23 and 0.34 on two bins. The two proton-tagged regions meet the stricter 0.68 criterion (0.74/0.75 in FHC and RHC). The proton-tagged version is extracted because it belongs to the exclusive final state, not because it is required to make p_π measurable.

6.2.13 The least-populated bin.

The RHC first bin unfolds to 0.41 times its A_C-smeared truth (0.472 ± 0.801 against 1.152) with a closure χ^2 of 1.23/2: a 0.8σ fluctuation on a bin whose uncertainty is twice its content. Independent throws move it by more than $\pm 1\sigma$. The bin holds too few events to constrain the



(a) FHC (b) RHC (c) Combined

Figure 4: The two-bin p_π cross section on the proton-tagged `CC1mu1pi1p` selection, the graphical form of Tab. 6. The first bin is that of the inclusive result of Fig. 2, and the second is the union of the two upper inclusive regions, since three regions fail the 0.50 criterion in the proton-tagged analysis sample. The bins are drawn equal-width.

cross section, and any real-data value in it should be read with its uncertainty rather than its central value.

6.2.14 Choice of boundary for the proton-tagged sample.

The 0.205 GeV/c boundary was independently preferred by the migration-level edge scan run on the proton-tagged sample (`macros/ppi_binning_scan.C`, FHC; `logs/binning_scan/1p_fhc.txt`) as well as by the inclusive scan, so it is not simply inherited. It was not re-optimised through a separate universe, covariance and unfolding production for every candidate boundary, which would need its own universe construction per configuration ($\sim$ 1–2 h each). This is not expected to change the conclusion, since both bins clear the criterion with margin in all three proton-tagged configurations.

6.3 Survey of pion momentum estimators without a magnetic field

Four approaches were measured on truth-matched pion tracks. All are limited by the same mechanism: each measures what the pion did before it interacted.

estimator	$\langle$reco/true$\rangle$, top bin	RMS	note
range (analysis)	0.319	0.16	saturates near 0.34 GeV/c
calorimetric energy	0.383	0.31	also truncated, and 2$\times$ noisier
MCS	1.011	2.1	unbiased, unusably noisy on short tracks
golden-pion selection	0.327	0.18	no gain above 0.8 GeV/c

Summary of the estimator survey. Five approaches were measured (§§27 and 27): range, calorimetry, MCS, golden-pion tagging and a multivariate regression, together with recovery of the interaction products. Each is limited by the same fact, that a charged pion which interacts before stopping destroys the information a non-magnetised detector uses to infer its momentum, and that a substantial part of the energy leaves the detector unseen. The regression is the only one that gives a material improvement, removing the $W_{\pi p}$ bias, at the cost of model dependence and of the low-W region.

One avenue has not been tried and is independent of the pion's own reconstruction: for the proton-tagged sample, transverse kinematic imbalance constrains the pion transverse momentum from the muon and proton alone. It is limited by Fermi motion, of order 250 MeV/c, which is comparable to the present uncertainties rather than smaller, so it would help but would not change the character of the measurement. Without it, p_π is a coarse measurement in a few wide

regions at this exposure and no differential $W_{\pi p}$ result is reported, both because of physics rather than of analysis choices.

7 Binning review against the 0.68 and 0.50 migration criteria

[SHARED]

The review established the rule and the binnings of §2. It starts from the binnings produced by the first optimisation and its re-check (§23), which are called the released binnings below and in the tables.

7.1 Criteria and selection rule

Admissibility criteria. A true bin is admissible when most of its selected signal is reconstructed in the same bin: the diagonal of the column-normalised migration matrix, measured on the analysis sample (§6.1), must reach 0.68, the value used in the BNB CC1π measurements, or 0.50, the relaxed criterion adopted for this analysis. The criterion bounds how much of a bin's content the unfolding moves between bins, and therefore how strongly the result leans on the modelled migrations. It is applied in FHC and in RHC; the combined configuration inherits the binning. Every true bin must also hold a minimum of selected signal events, counted in FHC at data exposure: 70 in the inclusive family, the content of the sparsest released inclusive bin, and 50 in the proton-tagged family. Neither requirement says how much of the binned shape survives the regularised unfolding, and that is the third property reviewed.

Retained shape: the Wiener filter factors. In the Wiener-SVD unfolding, A_C is similar to the diagonal matrix of the Wiener filter factors $W_k = S_k^2/(S_k^2 + N_k^2)$, one for each component of the pre-whitened response. Its eigenvalues are these factors, all between zero and one, and its trace is their sum: the number of degrees of freedom that the data determine. A factor of at least 0.5 marks a component measured with a signal-to-noise ratio of at least one. For the released FHC $\cos\theta_\mu$, for example, the factors are 0.79, 0.79, 0.66, 0.35 and 0.20, a trace of 2.79 in five bins. The first optimisation and its re-check (§23, Appendix B of the analysis note) used the conditioning $s_{\min}/s_1$ instead. The conditioning is set by the weakest component alone, so it does not separate a binning that loses one component from one that retains nothing beyond the normalisation, and the threshold of about 0.1 used there to reject finer candidates was not applied to the released binnings, most of which fall below it: p_μ (0.006 in FHC, 0.0001 in RHC), $\theta_{\mu\pi}$ (0.007, 0.002), θ_μ (0.06, 0.005) and RHC $\cos\theta_\mu$ (0.0003). The singular values of A_C are not the filter factors either: A_C is not symmetric, and its largest singular value exceeds one.

Candidates. For each observable and each number of bins up to twelve, the partition with the largest worst-bin diagonal over FHC and RHC that meets the event floor was found by dynamic programming on a fine grid, with the edge conventions of the released configurations (`report/tools/maximin_binning.py`). The tool reproduces the unfolder's diagonals of the released binnings to 0.01, and to 0.05 for W_{had}, two of whose edges fall between grid points. The partitions relevant to either criterion, from the coarsest that could replace the released binning to the finest passing 0.50, and for the angles also the partition at the released five bins, which separates the effect of the edges from that of the bin count, were extracted in full in FHC, RHC and combined: 141 extractions with universes, covariance, unfolding and closure , and the proton momentum p_p added to the proton-tagged family in two to five bins: 153 extractions in all (`data_release/binreview_eval_2026-09-26.tsv`). They carry the tune and pseudo-data predictions only. A partition has the same diagonals in $\cos\theta_\mu$ and in θ_μ, and the θ_μ grid is finer at forward angles, so the eight- and ten-bin θ_μ partitions, written in $\cos\theta_\mu$, are $\cos\theta_\mu$ candidates as well.

Selection rule. Among the admissible binnings of an observable, the one selected has the fewest bins that reach 90% of the largest trace of A_C, averaged over FHC and RHC; ties go to the larger worst diagonal. Bins beyond that point add components that the regularisation fills rather than the data. A choice is marked rule-sensitive when 85% or 95%, or an average that includes the combined configuration, would select a different binning. Figure 5 shows the trace against the number of bins for every candidate, Tables 7 and 8 the selected binnings, and Table 9 their edges (`report/tools/binreview_tables.py`, `data_release/binreview_choices_2026-09-26.tsv`).

7.2 Consequences and adopted binnings

Consequences of the two criteria. The released binnings fall into three groups. The seven-bin p_μ and the six-bin W_{had} fail both criteria, and the θ_p and $\theta_{\pi p}$ study binnings fall below the event floor: these change whichever criterion is adopted. The five-bin $\cos\theta_\mu$ and θ_μ and the two-bin $\delta\alpha_T$ fail 0.68 only; rebinning the muon angles to meet it gains information, while $\delta\alpha_T$ loses a second component whose filter factor is only 0.2. The remaining binnings, p_π, $\cos\theta_\pi$, $\theta_{\mu\pi}$, δp_T, $\delta\phi_T$ and p_n, pass both; of these $\cos\theta_\pi$ gains 0.45 degrees of freedom at seven bins under either criterion, and the others are at or near their information limit.

Where the two criteria select different binnings, the relaxed one adds 0.2 to 0.6 degrees of freedom per beam: p_μ (six bins against four), p_π (three against two), $\cos\theta_\mu$ (eight against seven), θ_μ (ten against eight), $\delta\phi_T$ (three against two) and $\theta_{\pi p}$ (four against three); for p_p (three against two) the gain is 0.15/0.12. $\delta\alpha_T$ and $W_{\pi p}$ gain a second bin whose component the data do not determine. A new component measured at signal-to-noise above one appears only for p_π in RHC, θ_μ and $\delta\phi_T$ in the combined configuration (where θ_μ loses one in RHC) and $\theta_{\pi p}$ in FHC and RHC. The price is diagonals down to 0.50: bins in which up to half of the selected events come from neighbouring true bins, with a correspondingly stronger dependence of the result on the modelled migrations.

Choice of criterion. The 0.68 criterion is that of the BNB measurements. Applied to every observable, it would end the exemption of p_μ, which rests on the regularisation absorbing migrations that the criterion exists to limit. Where it differs from 0.50 it forgoes at most 0.64 degrees of freedom per beam, and in all but five observable and beam combinations no component that the data determine at signal-to-noise above one; its cost falls on p_μ, which would drop from seven bins to four and from three well-measured components to two. The 0.50 criterion also ends the exemption. It keeps six p_μ bins, with two well-measured components as well, three p_π regions and eight $\cos\theta_\mu$ bins, and it accepts bins in which up to half of the selected events come from neighbouring true bins. The analysis uses the 0.50 criterion for both families. The stronger dependence on the modelled migrations that it admits is tested by the alternative-model closure and the ensembles, both repeated on the adopted binnings (Sec. 13.2 of the analysis note; Sec. 13.3 of the analysis note).

Adopted binnings. The adopted binnings are those the rule selects under 0.50 (Table 9), with four exceptions. $\cos\theta_\pi$ keeps its five bins, which pass both criteria (0.74/0.76); the seven bins the rule selects would add 0.44/0.46 degrees of freedom. $\theta_{\mu\pi}$ keeps its five bins (0.79/0.79), which retain as much as the six selected in FHC and 0.2 less in RHC. $\delta\alpha_T$ keeps its split at 120°: the rule's 122.5° raises the worst diagonal from 0.66 to 0.67 and changes the trace by less than 0.02. The proton-tagged p_π keeps two regions, because the three-region binning fails 0.50 in the proton-tagged sample. θ_μ is not reported: a partition has the same diagonals in θ_μ as in $\cos\theta_\mu$, and the cosine retains up to one degree of freedom more, because the regularisation acts on the density per unit of the variable (§24). The binnings are therefore: p_μ six bins and $\cos\theta_\mu$ eight in both families; p_π three regions in the inclusive family and two in the proton-tagged one; $\cos\theta_\pi$ five and four; $\theta_{\mu\pi}$ five; W_{had} two regions split at 1.18 GeV/c^2; $\delta\phi_T$ three bins; δp_T, $\delta\alpha_T$ and p_n

two; θ_p and $\theta_{\pi p}$ four at their maximin edges; and p_p three. No differential $W_{\pi p}$ result is reported. Table 1 and Table 1 of the proton-tagged note list the edges.

For every replaced binning the generator predictions were computed at the new edges from the event-level generator files, and the universes, covariances and result files were rebuilt (§22); the ensembles, the alternative-model closure and the D'Agostini cross-check were repeated on the adopted binnings. The control-region binnings are frozen separately (§20.5) and do not change.

7.3 Evidence for each observable

The worst diagonals of Tables 7 and 8 are measured on the analysis sample (§6.1).

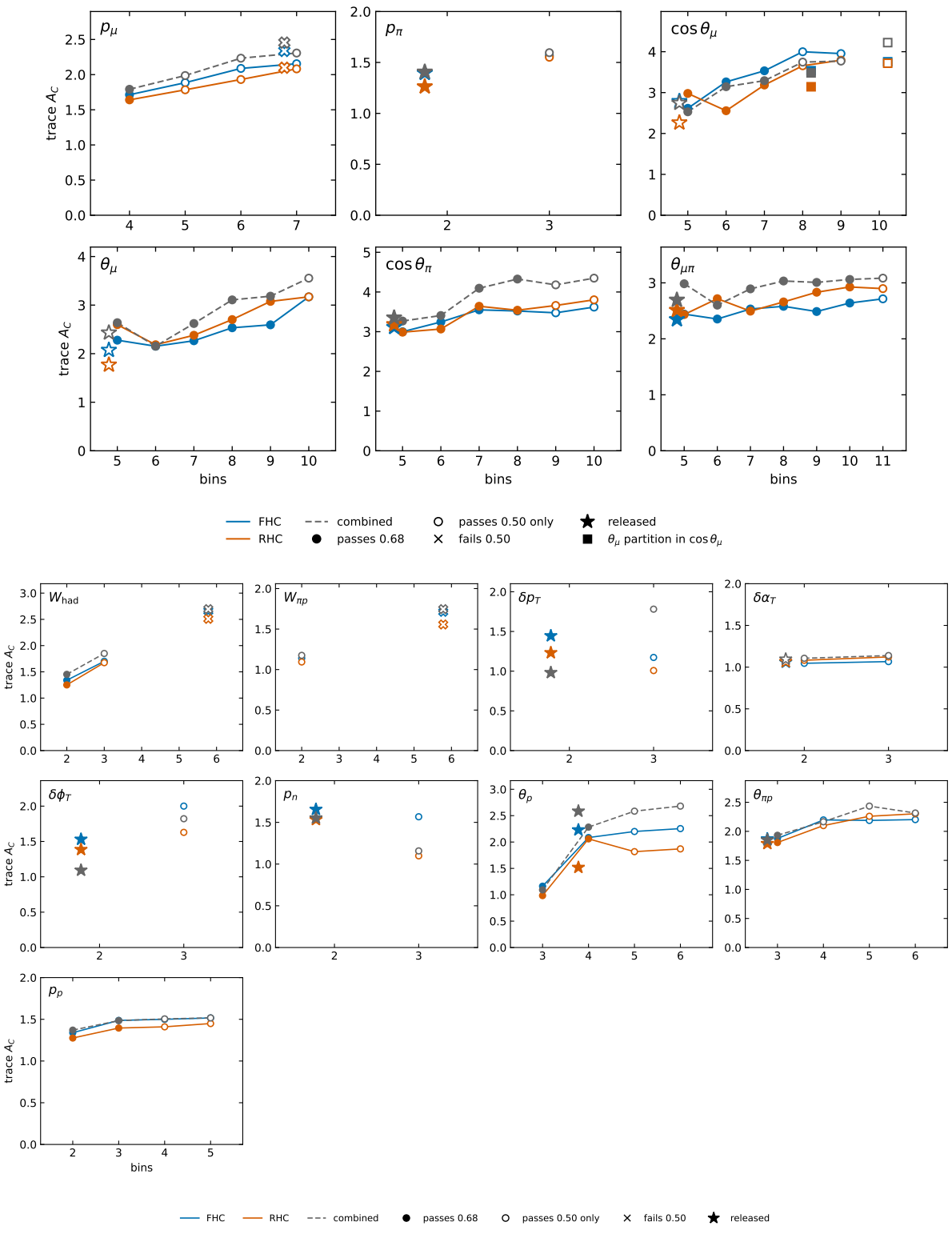


Figure 5: Degrees of freedom for signal (trace of A_C) against the number of bins, inclusive (top) and proton-tagged (bottom), FHC, RHC and combined. Lines join the maximin binnings; the star is the released binning (for θ_p and $\theta_{\pi p}$ the study binning), drawn just left of its bin count; squares are the θ_μ partitions written in $\cos\theta_\mu$. A filled marker passes 0.68 in FHC and RHC, an open one 0.50 only, a cross neither. The three-bin W_{had} is built at an event floor of 30, below that of the family. Pseudo-data; analysis sample.

Table 7: Inclusive binnings under the two criteria, FHC, RHC and combined. For each observable: the released binning and the binning the rule of §7 selects under the 0.68 and the 0.50 criterion. "worst diag." is the smallest column-normalised migration diagonal over the true bins; "trace A_C" the degrees of freedom for signal; "factors ≥ 0.5" the number of Wiener filter factors of at least 0.5, the shape components the data determine with signal-to-noise of at least one; the last column the largest per-bin total uncertainty. † The choice depends on the details of the rule (§7). Fake data.

	bins	worst diag. FHC/RHC	trace A_C FHC/RHC/COMB	factors ≥ 0.5 FHC/RHC/COMB	largest bin unc. [%] FHC/RHC/COMB
p_μ					
released	7	0.33/0.31	2.3/2.1/2.5	3/3/3	58/48/48
0.68	4	0.71/0.71	1.7/1.6/1.8	2/2/2	47/44/48
0.50†	6	0.57/0.59	2.1/1.9/2.2	2/2/2	47/49/43
p_π					
released	2	0.76/0.77	1.4/1.3/1.4	2/1/2	111/62/71
0.68 (= released)	2	0.76/0.77	1.4/1.3/1.4	2/1/2	111/62/71
0.50†	3	0.51/0.53	1.6/1.6/1.6	2/2/2	75/45/53
$\cos\theta_\mu$					
released	5	0.67/0.68	2.8/2.3/2.7	3/2/4	52/103/49
0.68†	7	0.70/0.70	3.5/3.2/3.3	4/4/4	54/61/53
0.50†	8	0.66/0.66	4.0/3.7/3.7	4/4/4	62/56/49
θ_μ					
released	5	0.66/0.65	2.1/1.8/2.4	2/2/3	33/46/44
0.68†	8	0.70/0.70	2.5/2.7/3.1	3/3/3	51/44/44
0.50†	10	0.64/0.66	3.2/3.2/3.6	3/2/4	45/60/65
$\cos\theta_\pi$					
released	5	0.74/0.76	3.1/3.2/3.4	4/3/4	56/48/50
0.68†	7	0.73/0.74	3.6/3.6/4.1	4/3/5	60/62/44
0.50†	7	0.73/0.74	3.6/3.6/4.1	4/3/5	60/62/44
$\theta_{\mu\pi}$					
released	5	0.79/0.79	2.3/2.5/2.7	3/3/4	45/39/39
0.68†	6	0.83/0.83	2.4/2.7/2.6	3/3/3	37/48/35
0.50†	6	0.83/0.83	2.4/2.7/2.6	3/3/3	37/48/35

Table 8: Proton-tagged binnings under the two criteria, as Table 7, with an event floor of 50 selected FHC signal events per bin. "study" marks the proposed θ_p and $\theta_{\pi p}$ binnings, which are not in the released result set. * Below the event floor: the released W_{had} and $W_{\pi p}$ hold 3 and 45 events in their sparsest bin, the study θ_p and $\theta_{\pi p}$ 21 and 42. † The choice depends on the details of the rule. Fake data.

	bins	worst diag. FHC/RHC	trace A_C FHC/RHC/COMB	factors ≥ 0.5 FHC/RHC/COMB	largest bin unc. [%] FHC/RHC/COMB
W_{had}					
released*	6	0.09/0.08	2.7/2.5/2.7	4/4/4	305/162/654
0.68	2	0.71/0.72	1.3/1.3/1.5	1/1/2	68/100/91
0.50	2	0.71/0.72	1.3/1.3/1.5	1/1/2	68/100/91
$W_{\pi p}$					
released*	6	0.03/0.03	1.7/1.6/1.7	2/1/2	496/145/122
0.68	1	no binning of two or more bins is admissible			
0.50	2	0.50/0.51	1.1/1.1/1.2	1/1/1	91/136/113
δp_T					
released	2	0.74/0.72	1.4/1.2/1.0	2/1/1	121/85/24
0.68 (= released)	2	0.74/0.72	1.4/1.2/1.0	2/1/1	121/85/24
0.50 (= released)†	2	0.74/0.72	1.4/1.2/1.0	2/1/1	121/85/24
$\delta\alpha_T$					
released	2	0.66/0.66	1.1/1.1/1.1	1/1/1	45/47/46
0.68	1	no binning of two or more bins is admissible			
0.50	2	0.67/0.66	1.0/1.1/1.1	1/1/1	46/49/46
$\delta\phi_T$					
released	2	0.79/0.76	1.5/1.4/1.1	2/2/1	189/152/37
0.68 (= released)	2	0.79/0.76	1.5/1.4/1.1	2/2/1	189/152/37
0.50	3	0.62/0.60	2.0/1.6/1.8	2/2/2	188/69/48
p_n					
released	2	0.77/0.75	1.7/1.5/1.6	2/2/2	172/222/155
0.68 (= released)	2	0.77/0.75	1.7/1.5/1.6	2/2/2	172/222/155
0.50 (= released)	2	0.77/0.75	1.7/1.5/1.6	2/2/2	172/222/155
θ_p					
study*	4	0.74/0.73	2.2/1.5/2.6	2/1/3	79/91/101
0.68	4	0.69/0.68	2.1/2.1/2.3	2/2/3	41/84/43
0.50†	4	0.69/0.68	2.1/2.1/2.3	2/2/3	41/84/43
$\theta_{\pi p}$					
study*	3	0.77/0.75	1.9/1.8/1.9	2/2/2	140/187/109
0.68	3	0.75/0.74	1.9/1.8/1.9	2/2/2	97/121/99
0.50†	4	0.67/0.67	2.2/2.1/2.2	3/3/2	94/141/60
p_p					
0.68†	2	0.87/0.86	1.3/1.3/1.4	2/1/2	48/61/52
0.50†	3	0.74/0.73	1.5/1.4/1.5	2/2/2	48/58/51

Table 9: Bin edges of the binnings selected under each criterion (Tables 7 and 8); the last bin of p_μ, p_π, W_{had}, $W_{\pi p}$, δp_T and p_n is open above, as in the release. "released" where the rule keeps the released edges.

Observable	criterion	edges
p_μ (GeV/c)	0.68	0.15, 0.3, 0.525, 0.95, 3
	0.50	0.15, 0.3, 0.425, 0.575, 0.825, 1.25, 3
p_π (GeV/c)	0.68	released
	0.50	0.175, 0.205, 0.26, 1
$\cos\theta_\mu$	0.68	-1, 0.55, 0.725, 0.825, 0.9, 0.95, 0.975, 1
	0.50	-1, 0.4, 0.625, 0.75, 0.825, 0.9, 0.95, 0.975, 1
θ_μ (rad)	0.68	0, 0.16, 0.239, 0.339, 0.439, 0.578, 0.738, 0.957, 3.15
	0.50	0, 0.16, 0.239, 0.319, 0.399, 0.498, 0.618, 0.738, 0.897, 1.156, 3.15
$\cos\theta_\pi$	0.68	-1, -0.15, 0.15, 0.375, 0.6, 0.775, 0.925, 1
	0.50	-1, -0.15, 0.15, 0.375, 0.6, 0.775, 0.925, 1
$\theta_{\mu\pi}$ (rad)	0.68	0, 0.55, 0.875, 1.225, 1.625, 2.05, 2.6
	0.50	0, 0.55, 0.875, 1.225, 1.625, 2.05, 2.6
W_{had} (GeV/c^2)	0.68	0, 1.18, 2.74
	0.50	0, 1.18, 2.74
$W_{\pi p}$ (GeV/c^2)	0.68	one bin
	0.50	1.08, 1.2, 2.9
δp_T (GeV/c)	0.68	released
	0.50	released
$\delta\alpha_T$ (deg)	0.68	one bin
	0.50	0, 122.5, 180
$\delta\phi_T$ (deg)	0.68	released
	0.50	0, 22.5, 75, 180
p_n (GeV/c)	0.68	released
	0.50	released
θ_p (rad)	0.68	0, 0.44, 0.785, 1.162, π
	0.50	0, 0.44, 0.785, 1.162, π
$\theta_{\pi p}$ (rad)	0.68	0, 1.131, 1.916, π
	0.50	0, 0.911, 1.414, 1.916, π
p_p (GeV/c)	0.68	0.3, 0.42, 1.5
	0.50	0.3, 0.42, 0.6, 1.5

Inclusive observables. p_μ. The released seven bins fail both criteria. Their FHC diagonals fall from 0.80 to 0.33 with momentum, because the bins are narrower than the 30% resolution of the MCS estimator; for that reason the released p_μ binning was exempted from the criterion (Sec. 11.6 of the analysis note). Under 0.68 four bins are admissible (worst diagonal 0.71 in both beams): the trace falls from 2.3/2.1 to 1.7/1.6, and the number of components measured with signal-to-noise above one from three to two. Under 0.50 the rule selects six bins (0.57/0.59, trace 2.1/1.9). Seven bins (0.51/0.51) add 0.1 and fail 0.50 in the proton-tagged sample, which uses the same edges (0.46/0.49), whereas the six-bin edges pass there (0.53/0.55). No admissible binning keeps the third well-measured component of the released one ($W = 0.62/0.52$); the closest is 0.40/0.46, at seven bins.

p_π. The released edges (0.175 and 0.205 GeV/c, open above) are the maximin two-bin partition, and no finer binning passes 0.68: the reconstructed momentum saturates near 0.34 GeV/c (§6.2). Under 0.50 a third bin from 0.26 GeV/c passes by 0.007 (0.51/0.53). Its own filter factor is small (0.05/0.01), but the split raises the second factor from 0.55/0.40 to 0.68/0.71, so the trace rises from 1.4/1.3 to 1.6/1.6 and the largest per-bin uncertainty falls from 111% to 75% in FHC. The same edges fail 0.50 in the proton-tagged sample (0.48 in FHC).

$\cos\theta_\mu$. The released five bins fail 0.68 in the backward bin (0.67/0.68). In RHC their fifth filter factor is 0.001 and the smallest A_C row sum 0.05, so one bin is filled almost entirely by the regularisation. The maximin five bins place four bins above $\cos\theta_\mu = 0.7$, where the events are, and pass at 0.757/0.765 with a trace of 2.6/3.0 and row sums of at least 0.61. Under 0.68 the rule

selects seven bins (0.70/0.70; trace 3.5/3.2, four components above signal-to-noise one in every configuration), and under 0.50 eight (0.66/0.66; trace 4.0/3.7), beyond which the trace levels off (nine bins: 3.95/3.79). The θ_μ partitions add one bin to each without adding information: eight bins pass 0.68 (0.70/0.69) with a trace of 3.5/3.1, and ten bins pass 0.50 (0.64/0.66) with 3.75/3.72.

θ_μ. This is the same variable in another parameterisation. The released binning fails 0.68 (0.66/0.65); the rule selects eight bins under 0.68 (0.70/0.70, trace 2.5/2.7) and ten under 0.50 (0.64/0.66, trace 3.2/3.2). The same partition retains less as θ_μ than as $\cos\theta_\mu$: the eight-bin partition keeps a trace of 2.5/2.7/3.1 as θ_μ and 3.5/3.1/3.5 as $\cos\theta_\mu$, the ten-bin one 3.2/3.2/3.6 and 3.75/3.72/4.22, with the same diagonals up to the three-decimal rounding of the edges. The regulariser penalises the curvature of the cross-section density, and $d\sigma/d\theta_\mu$ and $d\sigma/d\cos\theta_\mu$ are different functions, so the parameterisation is part of the regularisation. A θ_μ result adds no information to a $\cos\theta_\mu$ result with the same partition, and retains less.

$\cos\theta_\pi$. The released five bins pass both criteria (0.74/0.76). The trace rises to 3.6/3.6 at seven bins (0.73/0.74) and no further (eight to ten bins: 3.5–3.8), so both criteria select seven bins, with a largest per-bin uncertainty of 60/62% against 56/48%. The gain, 0.45 degrees of freedom in each beam and 0.7 combined, is the one improvement of the review that does not repair a criterion failure. The seven-bin edges give 0.69/0.71 and a bin of 30 events in the proton-tagged sample, whose $\cos\theta_\pi$ keeps its own four bins.

$\theta_{\mu\pi}$. The released five bins pass both criteria (0.79/0.79), and the trace barely moves between five and eleven bins (2.3–2.7 in FHC, 2.4–2.9 in RHC); no binning has more than four filter factors above 0.2. The observable carries about two and a half degrees of freedom at this exposure however it is binned. The rule selects six bins, rule-sensitively, and any count from five to ten is equivalent within its resolution. At five bins the maximin edges (0, 0.6, 1.025, 1.5, 2.0, 2.6 rad) raise the worst diagonal to 0.86 and the smallest row sum from 0.52/0.58 to 0.73/0.62 at unchanged trace.

Proton-tagged observables. W_{had}. The released six bins fail both criteria: their FHC diagonals are 0.23, 0.45, 0.57, 0.44, 0.22 and 0.09, below 0.68 in every bin and below 0.50 in five of the six. The first bin holds three selected signal events, the per-bin uncertainty reaches 305% in FHC, 162% in RHC and 654% combined, and the single-throw closure ratio of one combined bin is 78. Under either criterion the admissible binning is two regions split at 1.18 GeV/c^2 (0.71/0.72), with a trace of 1.3/1.3 and, in FHC and RHC, one component above signal-to-noise one. Three bins pass 0.50 only below the event floor (30 events: 0.531/0.535, per-bin uncertainty up to 325%).

$W_{\pi p}$. No two-bin partition passes 0.68. Under 0.50 a split at 1.20 GeV/c^2 passes by 0.003, with a second filter factor of 0.29/0.27: a normalisation, not a shape. Under either criterion no differential $W_{\pi p}$ result is reported.

δp_T *and* p_n. The released two bins pass both criteria (0.74/0.72 and 0.77/0.75). The three-bin maximin partitions pass 0.50 but lower the trace in FHC and RHC (δp_T: 1.45/1.23 to 1.17/1.01; p_n: 1.66/1.53 to 1.57/1.10), so both criteria keep two bins. In the combined configuration the two-bin δp_T has a trace of 0.98: its shape component is not measured there.

$\delta\alpha_T$. The released two bins fail 0.68 (0.66/0.66), and so does the best two-bin partition (0.67/0.66): under 0.68 there is no $\delta\alpha_T$ shape. Under 0.50 two bins are admissible, but their second filter factor is 0.19–0.24 in every configuration and a third bin adds nothing, so $\delta\alpha_T$ contributes one well-measured number under either criterion.

$\delta\phi_T$. The released two bins pass both criteria (0.79/0.76). Under 0.50 three bins (0.62/0.60; edges 0, 22.5, 75 and 180°) raise the trace from 1.5/1.4 to 2.0/1.6, and in the combined configuration from 1.1 to 1.8; the new factor is 0.40 in FHC but 0.01 in RHC.

θ_p. The study binning (0.74/0.73) holds 43 and 21 events in its two backward bins, below the floor, and its fourth filter factor in RHC is 0.001. The maximin four bins at the floor (0.69/0.68)

have a trace of 2.1/2.1 against 2.2/1.5, and a largest per-bin uncertainty of 41/84% against 79/91%; five and six bins add nothing. Both criteria select these four bins, which pass 0.68 by 0.002 in RHC. Three bins pass more comfortably (0.77/0.75) but retain a single component (trace 1.2/1.0), so they do not serve as a fallback.

$\theta_{\pi p}$. The last bin of the study binning holds 42 events, below the floor. Under 0.68 the maximin three bins (0, 1.131, 1.916 and π rad; 0.75/0.74) meet the floor and retain the same information as the study binning (trace 1.88/1.81 against 1.87/1.79) with a smaller largest per-bin uncertainty (97/121% against 140/187%). Under 0.50 four bins (0.67/0.67) raise the trace to 2.2/2.1 and add a third component above signal-to-noise one in FHC and RHC; five or six bins add nothing.

p_p. The momentum of the leading proton, measured from its range, is added to the family. The signal requires $p_p > 0.3$ GeV/c, which is the lower edge, and the upper bin is open. Two bins split at 0.42 GeV/c pass 0.68 (0.87/0.86), with filter factors 0.83 and 0.51 in FHC (0.80 and 0.48 in RHC). A third bin at 0.60 GeV/c keeps the same two components and measures the second better (0.68/0.64), at a worst diagonal of 0.74/0.73; its third factor is 0.001, and four or five bins add nothing (trace 1.49/1.40 at three bins, 1.52/1.45 at five). The rule selects the two bins under 0.68 and the three under 0.50.

Shared kinematic variables in the proton-tagged sample. The proton-tagged p_μ, p_π, $\cos\theta_\mu$ and $\theta_{\mu\pi}$ use the inclusive edges, evaluated here in the proton-tagged sample. The released p_μ gives 0.31/0.31 there and the released $\cos\theta_\mu$ 0.71/0.70. Of the selected inclusive binnings, the four-bin p_μ passes 0.68 by 0.005 in FHC (0.685/0.713), the six-bin p_μ passes 0.50 (0.53/0.55), the three-bin p_π fails 0.50 (0.48/0.51) and the five-bin maximin $\theta_{\mu\pi}$ passes at 0.84/0.86. The seven- and eight-bin $\cos\theta_\mu$ pass 0.68 (0.70/0.71) and 0.50 (0.65/0.68) there as well.

8 Hadronic mass [PROTON-TAGGED]

8.1 The W binnings are finer than the resolution

In the six-bin scheme, $W_{\pi p}$ shows the least model tension of the proton-tagged observables. The origin is the binning, not physics.

The six bin edges were chosen for equal population, without reference to the resolution. The resolution and the bin widths, measured on selected signal in FHC Runs 1/2/4 (analysis sample), are:

	$W_{\pi p}$	W_{had}
residual RMS [GeV/c^2]	0.142	0.283
robust half-width (16–84%)	0.101	0.177
bias (mean reco − true)	**−0.111**	−0.032
narrowest analysis bin	**0.040**	0.100
widest analysis bin	1.430	1.300
width ratio	**35.7**	13.0

The second and third $W_{\pi p}$ bins are 40 MeV/c^2 wide against a resolution of 101–142 MeV/c^2: they are two to three times narrower than the detector can resolve. The bias alone, −111 MeV/c^2, is larger than three of the six bin widths. The migration diagonals are correspondingly low, 87.6/22.3/11.7/12.0/10.0/2.3%: only the first bin is measured, and the top bin, at 2.3%, is largely unmeasured.

The five-bin p_π scheme fails in the same way and for the same reason. $W_{\pi p}$ is reconstructed from the pion and proton momenta, so it inherits the saturation of the range-based pion momentum described in §6.2; its resolution cannot be better than that of its inputs.

An exhaustive edge scan on the analysis sample, with the procedure and criterion used for p_π, gives:

scheme	diagonals	verdict
current, six bins	87.6/22.3/11.7/12.0/10.0/2.3%	1 of 6 usable
best two bins (split 1.15)	**83.2**/**77.9**%	**both usable**
best three bins	83.2/49.7/45.9%	fails
best four bins	83.2/45.0/28.2/28.2%	fails

As with p_π, two bins is the only scheme that satisfies the criterion, and its first bin is thin (137 selected signal events, against a floor of 100). For the model comparison this means that a low generator χ^2 at six bins reflects a binning that cannot resolve the differences between generators, not agreement between them. The same test applied to W_{had}, whose six-bin upper bins are also narrower than its 177 MeV/c^2 half-width, finds the six bins below both the 0.68 and the 0.50 criterion, and two regions split at 1.18 GeV/c^2 as the only admissible binning (§7); that is the binning used for W_{had}.

Two-bin scheme split at 1.15 GeV/c^2. The boundary is more tightly constrained than for p_π, where any value between roughly 0.19 and 0.22 satisfies the criterion. Here only one choice satisfies both the diagonal criterion and the 100-event floor, and it is the scheme extracted in §8.2:

boundary [GeV/c^2]	bin 1	bin 2	N(bin 1)	
1.13	76.5%	88.0%	51	below the event floor
1.15	**83.2**%	**77.9**%	**137**	**adopted**
1.17	86.8%	66.8%	266	bin 2 just fails
1.19	87.6%	56.0%	468	fails
1.23	91.7%	37.6%	1114	fails

Raising the boundary increases the first-bin population at the expense of the second-bin diagonal: the two diagonals trade against each other steeply, and the window in which both exceed 0.68 is only about ±0.01 GeV/c^2 wide. The window is this narrow because the resolution is comparable to the full width of the peak region.

The first bin is thin, 137 selected signal events against a floor of 100, so its uncertainty is large once systematics are included; the p_π first bin, with 469 events, carries 83% (Table 6). A boundary at 1.17 doubles the first-bin population for a bin-2 diagonal of 66.8%, marginally below threshold.

The scheme is implemented in `configs/ccpi1p_Wpipr_bin_config_2bin.txt`, with the reasoning recorded in a companion `.README`. The extraction with this scheme (§8.2) meets the migration criterion, but the first bin, with about 137 selected signal events, is suppressed by the Wiener filter, which drives A_C towards rank one for statistical rather than algebraic reasons. Widening the first bin restores its population at the cost of the upper-bin diagonal, and no scanned edge satisfies both requirements at once.

The apparent $W_{\pi p}$ agreement at six bins therefore does not persist in a binning the detector can resolve. No differential $W_{\pi p}$ result is reported. The topology-level one-bin total cross section of the proton-tagged sample remains measurable, but it is not described as a $W_{\pi p}$ measurement. W_{had} carries the same limitation, since it depends on the same pion momentum.

8.2 Limits of the $W_{\pi p}$ extraction

The two-bin $W_{\pi p}$ scheme of §8.1 was run through the full extraction. It fails for a reason distinct from the six-bin failure, and the follow-up studies establish what the pion momentum can and cannot deliver.

The two-bin extraction fails on statistics. The response behaves as designed: the framework returns migration diagonals of 0.802 and 0.773, matching the truth-level scan, and the closure is acceptable at $\chi^2 = 1.17/2$ ($p = 0.56$). The extraction nonetheless carries almost no shape information, because A_C collapses. The matrices below come from the extraction pass in which the two-bin $W_{\pi p}$ scheme was run and are written with true bins as rows; the two p_π rows are the same-pass comparison, and the released inclusive p_π matrix is that of Fig. 3.

two-bin scheme	A_C	s_2/s_1
$W_{\pi p}$, first bin $N = 137$	$\begin{pmatrix} 0.018 & -0.410 \\ 0.034 & 0.892 \end{pmatrix}$	**0.031**
p_π proton-tagged, $N = 469$	$\begin{pmatrix} 0.498 & -0.089 \\ 0.013 & 0.854 \end{pmatrix}$	0.577
p_π inclusive, $N = 1056$	$\begin{pmatrix} 0.594 & -0.382 \\ 0.011 & 0.891 \end{pmatrix}$	0.528

The top-left entry is 0.018: A_C passes under two per cent of the first bin and takes -0.41 from the second. This is a rank-one collapse of the kind described in §24, but its cause is statistical rather than a defective regulariser. With 137 events the first bin has such poor signal-to-noise that the Wiener filter suppresses that mode almost entirely, and the ordering across the three schemes, 137 → 0.031, 469 → 0.577, 1056 → 0.528, follows population rather than anything geometric.

$W_{\pi p}$ is therefore subject to two incompatible requirements. Bins narrow enough to satisfy the response criterion leave the first bin too sparse to unfold; bins wide enough for a stable unfolding (1.19 gives 468 events, matching the working p_π case) fail the criterion at 56%.

No two- or three-bin scheme satisfies both requirements. A scan of every two- and three-bin scheme with edges between 1.12 and 1.70 GeV/c^2 (FHC Run1+2+4, 3505 selected signal events) finds none that keeps both a migration diagonal above 0.68 and adequate statistics in every bin:

two-bin edge	N_1/N_2	diag_1 / diag_2	
1.15	137/3367	0.832/0.779	first bin too sparse
1.18	351/3153	0.866/0.611	
1.24	1298/2206	0.930/0.347	
1.27 (median)	1744/1760	0.951/**0.261**	upper bin unusable
1.45	2966/538	0.993/0.030	

The trend follows from the reconstruction bias. Reconstructed $W_{\pi p}$ is biased about 111 MeV low, inheriting the saturation of the range-based pion momentum (§6.2), so raising the edge far enough to balance the statistics drains the upper bin by downward migration: diag_1 climbs from 0.786 to 0.999 across the scan while diag_2 collapses from 0.927 to 0.004. There is no edge at which both bins are simultaneously populated and diagonal, and no three-bin combination does better: the third-bin diagonal never exceeds 0.347.

At this exposure $W_{\pi p}$ does not support a shape measurement unless the pion momentum improves. No differential $W_{\pi p}$ result is reported. The topology-level one-bin total cross section of the proton-tagged sample remains measurable, but it is not described as a $W_{\pi p}$ measurement. It is a one-bin extraction rather than a regularised unfolding: with a single bin there is nothing to unfold, since the response is 1×1, A_C is trivially unity and the regularisation has no effect. It is reported through the total flux-averaged cross-section path of Sec. 14.6 of the analysis note. Generator predictions integrated over the $W_{\pi p}$ range are built by `generator_predictions/newg4/make_wpipr_1bin.C`, summing the per-bin cross sections of the six-bin flux-averaged predictions.

Six-bin $W_{\pi p}$ closure, diagnostic only. The six-bin extraction was carried through the full chain in all three configurations, detector variations included. It closes against the A_C-smeared

truth, but that closure cannot restore shape information the response does not contain (the top bin is 2% diagonal), so these numbers are a check of the machinery and not a result, and they do not appear in the result tables of the proton-tagged note. The six-bin generator predictions (GENIE 0.501, NuWro 0.646, GiBUU 0.735, NEUT 0.783 $\times 10^{-38}$ cm^2/Ar, FHC) are in the release for the same diagnostic purpose.

Table 10: Six-bin $W_{\pi p}$ pseudo-data closure, diagnostic only (not reported as a differential result). σ_{int} in 10^{-38} cm^2/Ar.

Config	σ_{int}	unf./truth	χ^2/ndf
FHC	0.471	0.98	1.32/6
RHC	0.318	0.84	2.09/6
Combined	0.360	0.84	0.56/6

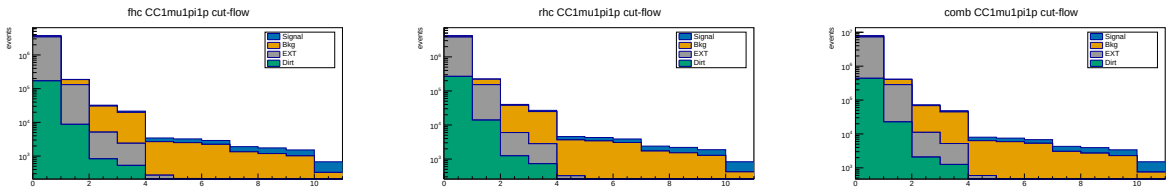
The limitation comes from the pion momentum alone. Correlating the $W_{\pi p}$ residual with each input residual isolates the cause:

input	r	W residual RMS when…	
pion momentum	**+0.726**	all events	0.142
proton momentum	+0.012	pion momentum well reconstructed	**0.061**
pion $\cos\theta$	−0.029	proton momentum well reconstructed	0.130
proton $\cos\theta$	−0.032	both well reconstructed	0.038

Fixing the pion momentum alone takes the resolution from 0.142 to 0.061, a factor 2.3; fixing the proton momentum changes little, and the angles have no effect. The hadronic interaction that truncates the pion track therefore limits p_π to two bins and excludes a shape measurement of $W_{\pi p}$.

Reparameterisation to W^2 leaves the migration unchanged. A monotonic transformation with mapped edges leaves the migration matrix unchanged, and this was verified directly: splitting at 1.15 GeV/c^2 in W and at $1.15^2 = 1.3225$ GeV$^2/c^4$ in W^2 gives diagonals of 83.2% and 77.9% in both cases, on identical events. Resolution and bin width scale together, $\sigma(W^2) = 2W\sigma(W)$, so σ/width is invariant at about 2 either way. The same argument applies to θ_μ versus $\cos\theta_\mu$ (§24).

9 Proton-tagged selection, binning and additional smearing [PROTON-TAGGED]



(a) FHC. (b) RHC. (c) Combined.

Figure 6: Proton-tagged (`CC1mu1pi1p`) cut-flow yields at full exposure (log scale; MC-prediction stack, Okabe-Ito: signal blue, background orange, EXT grey, dirt green). The last stage, IdentProton, is the leading-proton identification that defines the proton-tagged sample.

Table 11: Proton-tagged (`CC1mu1pi1p`) cut-flow yields at full exposure for FHC, RHC and combined (MC prediction; data blind). Stages None–Nonprotons are the inherited inclusive selection with the proton-tagged signal definition; the final IdentProton stage adds the leading-proton identification. Signal/Bkg from the ν_μ overlays POT-scaled per run, reprocessed with the proton-tagged selection and the full central-value weight under the safe-weight rule (`reprocess_cutflow_cf_1p.sh`); EXT from the per-run beam-off samples scaled by the run trigger ratios; dirt from the GENIE NuMI dirt overlay (`macros/cutflow_yields_1p.C`). The final row agrees with the proton-tagged cut-and-count check (Table 5 of the proton-tagged note). Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to the generated signal (the None stage, where all true signal is counted); Pur. = Signal/Pred.

Cut	Signal	Bkg MC	EXT	Dirt	Pred.	Eff. [%]	Pur. [%]
FHC (Runs 1,2,4,5; 7.766×10^{20} *POT)*							
None	2 888	237 145	3 256 689	170 169	3 666 891	100.0	0.08
InFiducialVol	2 424	54 365	122 860	8 738	188 387	83.9	1.3
Topological	1 917	25 340	4 299	827	32 383	66.4	5.9
MuonCandidate	1 739	17 435	1 870	530	21 573	60.2	8.1
ContainedPion	762	2 411	179	91	3 444	26.4	22.1
MuonIn3Planes	726	2 289	149	67	3 232	25.1	22.5
PionIn3Planes	675	2 076	116	38	2 906	23.4	23.2
ShowerCut	565	1 222	86	32	1 905	19.6	29.7
OpeningAngle	561	1 145	33	10	1 749	19.4	32.1
Nonprotons	508	984	24	6.6	1 522	17.6	33.4
IdentProton	**349**	**319**	**6.9**	**2.1**	**677**	**12.1**	**51.5**
RHC (Runs 1,2,3,4a,4b,4c; 1.108×10^{21} *POT)*							
None	3 427	288 891	3 560 975	267 449	4 120 741	100.0	0.08
InFiducialVol	2 886	67 416	139 122	13 907	223 331	84.2	1.3
Topological	2 318	31 870	4 767	1 243	40 199	67.7	5.8
MuonCandidate	2 111	21 962	2 090	730	26 892	61.6	7.8
ContainedPion	924	3 325	195	128	4 572	27.0	20.2
MuonIn3Planes	882	3 152	162	100	4 295	25.7	20.5
PionIn3Planes	823	2 867	124	71	3 885	24.0	21.2
ShowerCut	679	1 572	88	59	2 398	19.8	28.3
OpeningAngle	675	1 472	33	18	2 198	19.7	30.7
Nonprotons	602	1 246	24	9.5	1 881	17.6	32.0
IdentProton	**413**	**411**	**7.3**	**1.8**	**834**	**12.1**	**49.6**
Combined FHC+RHC (1.885×10^{21} *POT)*							
None	6 315	526 036	6 817 663	437 617	7 787 632	100.0	0.08
InFiducialVol	5 310	121 781	261 981	22 645	411 718	84.1	1.3
Topological	4 235	57 210	9 066	2 071	72 582	67.1	5.8
MuonCandidate	3 849	39 396	3 960	1 259	48 465	61.0	7.9
ContainedPion	1 686	5 736	374	220	8 016	26.7	21.0
MuonIn3Planes	1 608	5 441	311	167	7 527	25.5	21.4
PionIn3Planes	1 498	4 943	241	109	6 791	23.7	22.1
ShowerCut	1 244	2 794	174	91	4 303	19.7	28.9
OpeningAngle	1 236	2 617	66	28	3 947	19.6	31.3
Nonprotons	1 110	2 230	48	16	3 404	17.6	32.6
IdentProton	**762**	**730**	**14**	**3.9**	**1 510**	**12.1**	**50.4**

Table 12: Binning of the nine proton-tagged observables specific to the family, read from the data release (`curves_1p_*.tsv`). Every binning except the six-bin $W_{\pi p}$ meets the 0.50 migration criterion on the analysis sample with at least 50 selected FHC signal events per bin (§7); the last bin of the momenta and masses is open at truth level (the upper edge listed is the plotting boundary). $W_{\pi p}$ brackets the $\Delta(1232)$ peak; no differential $W_{\pi p}$ result is reported. The five inherited kinematic observables (p_μ, p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$) use the binnings of Table 1 of the proton-tagged note.

Observable	Definition	Bins	Released bin edges
p_p	leading proton, range	3	0.3, 0.42, 0.6, 1.5 GeV/c
$W_{\pi p}$	$M(\pi p)$, direct	6 (Not reported)	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9 GeV/c^2
$W_{\rm had}$	calorimetric	2	0, 1.18, 2.74 GeV/c^2
δp_T	$\lvert\vec{p}_T^{\,\mu} + \vec{p}_T^{\,\rm had}\rvert$	2	0, 0.3, 2.5 GeV/c
$\delta\alpha_T$	TKI angle	2	0, 120, 180°
$\delta\phi_T$	TKI angle	3	0, 22.5, 75, 180°
p_n	inferred nucleon momentum	2	0, 0.375, 2 GeV/c
θ_p	proton polar angle	4†	0, 0.44, 0.785, 1.162, π rad
$\theta_{\pi p}$	π–p opening angle	4†	0, 0.911, 1.414, 1.916, π rad

† Maximin edges of the binning review.

9.1 Reconstruction of the proton-tagged observables from input quantities

All six observables are built from the three candidate tracks (muon, charged pion, proton) using only reconstructed quantities stored in the input ntuples: the track direction cosines $\hat{n}$, the muon-hypothesis range and multiple-scattering momenta and the proton-hypothesis range kinetic energy.

Candidate three-momenta. The muon momentum magnitude is range-based when the muon track ends inside the containment volume (Sec. 7.1 of the analysis note), and multiple-Coulomb-scattering based otherwise. The charged pion, being minimum-ionising, starts from the muon-hypothesis range momentum, converted to the pion hypothesis at fixed kinetic energy in the reconstructed-bin definition exactly as in the inclusive analysis (Sec. 8 of the analysis note); the same configuration formula is used. The proton momentum follows from its kinetic energy under the proton hypothesis, $p_p = \sqrt{T_p^2 + 2m_p T_p}$, with T_p taken from the track range (`trk_energy_proton_v`, an exact function of the track length). Each three-momentum is $\vec{p}_i = p_i\,\hat{n}_i$, and the energies are $E_\mu = \sqrt{p_\mu^2 + m_\mu^2}$, $E_\pi = \sqrt{p_\pi^2 + m_\pi^2}$, $E_p = T_p + m_p$ (masses $m_\mu = 105.7$, $m_\pi = 139.6$, $m_p = 938.3$ MeV).

Directly measured mass $W_{\pi p}$. The invariant mass of the pion–proton four-vector sum, using no neutrino-direction assumption:

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - \lvert\vec{p}_\pi + \vec{p}_p\rvert^2}.$$

Transverse kinematic imbalance. The TKI observables treat the muon against the visible hadronic system $\vec{p}_{\rm had} = \vec{p}_\pi + \vec{p}_p$, $E_{\rm had} = E_\pi + E_p$. With the components transverse to the neutrino direction (the $\hat{z}$ axis of the beam frame) written $\vec{p}_T^{\,\mu}$ and $\vec{p}_T^{\,\rm had}$,

$$\delta p_T = \lvert\vec{p}_T^{\,\mu} + \vec{p}_T^{\,\rm had}\rvert, \quad \delta\alpha_T = \arccos\frac{-\vec{p}_T^{\,\mu}\cdot(\vec{p}_T^{\,\mu} + \vec{p}_T^{\,\rm had})}{\lvert\vec{p}_T^{\,\mu}\rvert\,\delta p_T}, \quad \delta\phi_T = \arccos\frac{-\vec{p}_T^{\,\mu}\cdot\vec{p}_T^{\,\rm had}}{\lvert\vec{p}_T^{\,\mu}\rvert\,\lvert\vec{p}_T^{\,\rm had}\rvert},$$

both angles in degrees folded onto $[0, 180]$. For a stationary nucleon with no final-state interactions $\delta p_T \to 0$; Fermi motion broadens it and FSI drive $\delta\alpha_T \to 180°$.

Calorimetric mass $W_{\rm had}$ and struck-nucleon momentum p_n. The calorimetric available energy is $E_{\rm cal} = E_\mu + (E_{\rm had} - m_p) + E_b$ with the argon removal energy $E_b = 24.78$ MeV, giving the

momentum transfer $Q^2 = -\left[(0,0,E_{\rm cal},E_{\rm cal}) - p^\mu\right]^2$ and

$$W_{\rm had} = \sqrt{m_p^2 + 2m_p(E_{\rm cal} - E_\mu) - Q^2}$$

(returned as 0 where reconstruction fluctuations make the argument negative). The nucleon-momentum proxy p_n combines the transverse imbalance with a longitudinal component p_L from energy–momentum balance, $p_n = \sqrt{\delta p_T^2 + p_L^2}$, using the MINERvA prescription [1] as implemented in the framework's transverse-kinematics tool (option 1):

$$p_L = \frac{R}{2} - \frac{M_{A'}^2 + \delta p_T^2}{2R}, \qquad R = M_A + p_z^\mu + p_z^{\rm had} - E_\mu - E_{\rm had},$$

with the target mass $M_A = 22\,m_n + 18\,m_p - 0.34381$ GeV (argon-40), the residual mass $M_{A'} = M_A - m_n + E_b$ and $E_b = 24.78$ MeV; z is the beam axis (Sec. 4.2 of the analysis note). The tool was written for a single-proton hadronic system: it receives the summed three-momentum $\vec{p}_\pi + \vec{p}_p$ and energy $E_{\rm had} = E_\pi + E_p$ in the proton slot and forms the hadronic kinetic energy as $E_{\rm had} - m_p$, which is why m_p appears in $E_{\rm cal}$ and $W_{\rm had}$ above. δp_T, $\delta\alpha_T$ and $\delta\phi_T$ depend only on the transverse momenta and are exact for the $\mu + (\pi p)$ system; p_L, p_n and $W_{\rm had}$ inherit the single-nucleon residual-mass and removal-energy assumptions of a quasi-elastic-oriented prescription and are therefore proxies, not the initial-state nucleon momentum. The generator predictions apply the identical formulas to the generator final states (Sec. 9.2), so the comparison is like for like; a truth-level closure of this prescription against a pion-production-specific definition, and a quantification of the bias from the residual-mass assumption, have not been performed (Sec. 8 of the proton-tagged note).

Two exposure conventions in the processed files. One normalisation point is specific to the proton-tagged sample. The processed ntuples use two POT conventions: the files behind the inclusive selection store the total run MC POT duplicated in every file of a multi-file run, whereas the proton-tagged files store each file's own POT. Normalising by the individual file POT would be correct for the first set but would scale every split of a multi-file run to the full run exposure in the second, inflating the simulated background by the split multiplicity ($\sim$3$\times$ for RHC) and driving the background-subtracted spectrum negative; normalising by the summed per-file POT has the mirror problem. The analysis therefore normalises each run by the sum of the distinct per-file POT values within that run, which reduces to the correct run total under either convention; FHC, with one MC file per run, is unaffected by the choice.

9.2 Additional smearing in the proton-tagged observables

Because the Wiener–SVD additional-smearing matrix A_C is not norm-preserving, each observable's predictions are rescaled by a different factor when they are brought into the measurement space. Table 13 gives that factor, measured as the ratio of the A_C-smeared to the truth-level generator integral, for the proton-tagged observables. Its dominant behaviour is set by the response and the binning; the exact factor is prediction-dependent, and for the tested generators GENIE and GiBUU agree to a few per cent throughout.

In the released extractions the proton-tagged observables are suppressed by 0.50–0.85 (Table 13): the two-region $W_{\rm had}$ is the least affected (0.85), the two-bin TKI variables, $W_{\pi p}$, p_p and the proton angles sit at 0.65–0.77, and the three-bin $\delta\phi_T$ is the most affected (0.50). The suppression is not driven by sharpness alone: $W_{\pi p}$ is the sharpest distribution measured here (the ratio of its largest to smallest bin is $\sim$36) yet is suppressed no more than the TKI variables, because A_C removes structure that the response cannot resolve and $W_{\pi p}$ is built from directly measured pion and proton momenta with equal-population bins that track the $\Delta(1232)$ peak.

No observable is enhanced by A_C: in the released configurations every prediction-dependent integral factor is below unity. This is an empirical property of these extractions, not a mathematical requirement on A_C, which is a regularisation operator and can suppress or enhance; an amplification would signal a mis-specified regularisation (§24). The suppression is applied identically to data, tune and generators, so model comparisons read off any panel are made on equal footing.

Table 13: Mean A_C row sum for the proton-tagged observables (FHC), from the released matrices `A_C_1p_FHC5_*.tsv`; equal to the ratio of the A_C-smeared to the truth-level integral for a flat prediction.

† Secondary result; $W_{\pi p}$ is Not reported and the others are each a Validation-only product.

Observable	mean A_C row sum
W_{had}	0.85
$\delta\alpha_T$	0.77
θ_p†	0.76
p_n	0.75
$W_{\pi p}$	0.75
p_p	0.74
δp_T	0.67
$\theta_{\pi p}$†	0.65
$\delta\phi_T$	0.50

Computed from the released matrices (`A_C_1p_FHC5_*.tsv`). The values span 0.50–0.85*, and those of the inclusive FHC results* 0.59–0.90*, so differences in smearing cannot order the generator* χ^2 *across observables.*

10 Two-dimensional pairs [SHARED]

Eight two-dimensional pairs were screened and extracted in the three configurations. A two-dimensional binning is a flat list of analysis bins ordered slice by slice, so the extraction uses the same machinery, and the integrated normalisation of the two-dimensional chain agrees with the one-bin total to a median of 1.5% once each is compared with its own A_C-smeared truth. Two pairs are usable, both in the combined configuration only: $\theta_p \times \delta p_T$ ($s_{\min}/s_1 = 0.611$) and $\cos\theta_\pi \times \cos\theta_\mu$ (0.564) (Table 14). Combining the beam modes roughly doubles their conditioning ($0.43 \to 0.61$ and $0.25 \to 0.56$), through the larger sample and the shared flux rather than through any separation of ν and $\bar{\nu}$. The pairs built on $W_{\pi p}$ are not usable, as the 39% leakage between its outer bins implies.

The two usable pairs are each a Secondary result in the combined configuration only: $d^2\sigma/d\theta_p\, d\delta p_T$ on the proton-tagged sample, presented in the proton-tagged note, and $d^2\sigma/d\cos\theta_\pi\, d\cos\theta_\mu$ on the inclusive sample, presented in Sec. 14.7 of the analysis note. Their FHC and RHC extractions and the other six pairs are Not reported. The single-mode extractions of the two pairs are not shape measurements, because their responses cannot be inverted. Every one-dimensional result is quoted per beam mode and for the combination; the two two-dimensional results are quoted in the combined configuration only.

The pair $\theta_{\pi p} \times W_{\pi p}$ conditions at 0.537, as do the two usable pairs, yet its extraction carries no usable information: a 78% data-statistical term, per-bin uncertainties up to 276% and closure ratios down to -0.80. Conditioning bounds how much of the response can be inverted and says nothing about how much data survives in the inverted directions, so the two must be read together. Conversely, $\cos\theta_\mu \times p_\mu$ screened best of the eight on the migration diagonal (0.70/0.68,

analysis sample) and is rank-deficient in FHC (0.002), while $\theta_p \times \delta p_T$ screened worst of the proton-tagged pairs and conditions best of all. The migration diagonal is a necessary condition, not a sufficient one; the one-dimensional binning review measures what the regularised response retains by the Wiener filter factors of A_C (§7).

Table 14: The eight two-dimensional pairs (inner × outer), ordered by the conditioning of A_C in the combined configuration. The last three columns refer to the combined configuration: the bin-averaged data-statistical term, the range of the per-bin total uncertainty and the range of the closure ratio against the A_C-smeared truth. Pseudo-data.

	$s_{\min}/s_1$			combined configuration		
Pair	FHC	RHC	COMB	data stat.	unc. [%]	closure
$\theta_p \times \delta p_T$	0.433	0.332	**0.611**	13.3%	29–103	0.70–1.36
$\cos\theta_\pi \times \cos\theta_\mu$	0.245	0.315	**0.564**	14.9%	33–64	0.91–1.26
$\theta_{\pi p} \times W_{\pi p}$	0.379	0.214	0.537	78.3%	39–276	−0.80–1.22
$\cos\theta_\pi \times \theta_{\mu\pi}$	0.037	0.133	0.171	25.5%	29–93	0.96–1.29
$\cos\theta_\mu \times p_\mu$	0.002	0.261	0.106	9.0%	33–57	0.97–1.04
$\theta_p \times W_{\pi p}$	0.012	0.152	0.112	200.4%	36–568	0.23–1.03
$\cos\theta_\pi \times p_\pi$	0.024	0.158	0.052	13.8%	32–62	0.96–1.11
$\theta_{\pi p} \times \delta p_T$	0.023	0.035	0.027	10.4%	26–52	0.89–1.15

11 Bin widths in the regularisation operator [SHARED]

Wiener–SVD regularises with a matrix C; this analysis uses the second-derivative form, built on the density with the actual bin widths, because of what the plain tridiagonal stencil $(1, -2, 1)$ does on these binnings. That stencil is a second derivative only on a uniform grid; these binnings are strongly non-uniform, with width ratios up to 14:1 (Table 16), so on them it is not a second derivative at all. It is near-singular, its row sums are the 10^{-6} regularisation floor, and its near-null direction is the constant vector rather than anything physical. The Wiener filter is then free to retain that direction alone, which is a rank-1 A_C that passes normalisation and discards shape.

The FHC θ_μ extraction at its pre-review binning (§24) is the clearest case, because its bin widths span a factor of 10.4. Unfolded with the uniform stencil, the unfolded data, the A_C-smeared truth, the MicroBooNE tune and all four generators share the same normalised shape to four decimal places (Table 15): writing the four generator predictions as the columns of an input matrix and their A_C-smeared counterparts as the columns of the output matrix, the inputs are full rank ($s_2/s_1 = 5.0 \times 10^{-2}$) while the outputs give $s_2/s_1 = 2.8 \times 10^{-6}$. Every shape degree of freedom is annihilated, each model is mapped onto one common curve and only its normalisation survives; any agreement with the data is then true by construction, and a model comparison made on such an extraction is meaningless. $\cos\theta_\mu$, the next most non-uniform binning at 14.5, is the other observable strongly affected; the remaining observables sit at width ratios of 3–6 and are affected far less.

The operator used. Both derivative regularisation matrices difference the density on the actual grid: each column is divided by its bin width and the stencil uses the true spacings between bin centres, with a one-sided first-derivative stencil at the ends. (A Dirichlet closure was rejected: it asserts the distribution vanishes at the edges, which is false here, $\cos\theta_\mu$ peaks at its upper edge.) With no widths supplied the code reduces to the uniform stencil. The widths are passed in by the unfolding application, which holds the slice definitions that the cross-section extractor does not. The effect on the conditioning of A_C, measured as s_2/s_1 of the smearing matrix itself, is given in Table 16: every inclusive observable improves and none is degraded

Table 15: FHC θ_μ unfolded with a uniform-grid second-derivative stencil: normalised bin contents. Five independent inputs, the measurement and four generator predictions plus the tune, collapse onto one shape, the signature of a rank-1 A_C.

	bin 1	bin 2	bin 3	bin 4	bin 5
Unfolded data	0.1485	0.3361	0.2899	0.1934	0.0321
A_C-smeared truth	0.1484	0.3359	0.2901	0.1933	0.0323
MicroBooNE tune	0.1484	0.3359	0.2901	0.1934	0.0322
NEUT	0.1484	0.3359	0.2901	0.1934	0.0322

except combined $\cos\theta_\mu$, marginally; in the proton-tagged family combined $W_{\pi p}$ (0.022) and $\delta\phi_T$ (0.083) would be as compromised as FHC θ_μ under the uniform stencil.

Table 16: A_C conditioning s_2/s_1 with the uniform-grid stencil ("uniform") and with the width-aware operator ("width-aware"), measured on one pseudo-data pass at the binnings under test (the $\cos\theta_\pi$ row is a five-bin scheme other than the reported one); the width-aware values of the reported binnings follow from the released matrices. With the bin-width ratio of each observable. A value approaching zero means the additional-smearing matrix has collapsed onto a single mode and the extraction carries no shape information.

	width	FHC		RHC		COMB
observable	ratio	uniform	width-aware	uniform	width-aware	uniform → width-aware
p_μ	6.3	0.34	0.90	0.31	0.94	0.19 → 0.68
$\cos\theta_\mu$	14.5	0.46	0.57	0.45	0.55	0.42 → 0.39
$\cos\theta_\pi$	3.6	0.67	0.71	0.55	0.79	0.55 → 0.74
$\theta_{\mu\pi}$	3.0	0.42	0.85	0.47	0.65	0.42 → 0.46
θ_μ	10.4	**0.015**	**0.51**	0.22	0.62	**0.045 → 0.38**
p_π (2 bins)	–	0.46	0.53	0.41	0.44	0.29 → 0.40
proton-tagged, combined configuration						
$W_{\pi p}$	–					**0.022 → 0.60**
W_{had}	–					0.23 → 0.86
δp_T	–					0.25 → 0.93
$\delta\alpha_T$	–					0.24 → 0.48
$\delta\phi_T$	–					**0.083 → 0.85**
p_n	–					0.21 → 0.88

12 The D'Agostini cross-check [INCLUSIVE]

Seven unfolding methods, listed below, are run in parallel with the Wiener-SVD extraction. The D'Agostini comparison (Table 18 and the numbers around it) is evaluated on the released pseudo-data. The closure figures of the other methods were evaluated on a different pseudo-data throw; the comparison they support is a property of the operators and does not depend on the throw.

Table 17: Unfolding methods.

Label	Description
`wiener_svd`	Wiener-SVD (every reported result)
`bin_by_bin`	Efficiency correction only; no migration unfolding
`ibu`	D'Agostini iterative Bayesian, 4 iterations
`tikhonov`	Tikhonov-regularised least-squares, $\tau = 1.0$
`roo_bayes`	RooUnfoldBayes, 4 iterations
`roo_svd`	RooUnfoldSvd
`roo_invert`	RooUnfoldInvert (direct matrix inversion)
`roo_binbybin`	RooUnfoldBinByBin

The D'Agostini iterative-Bayesian method provides the principal quantitative cross-check of the Wiener-SVD result. Because it does not apply the Wiener-SVD additional-smearing matrix A_C, its unfolded total is free of the (non-norm-preserving) A_C suppression and is stable against the iteration count. Table 18 compares the two integrated totals for all three configurations: the D'Agostini result sits above the Wiener-SVD total by 30% (FHC), 30% (RHC) and 27% (Combined), equivalently A_C suppresses the quoted Wiener-SVD integral by 23%, 23% and 21%, and it varies by $\leq 0.3\%$ across 1, 2 and 4 iterations. The D'Agostini integral is flat across the five observables to 0.6% (FHC, 1.036–1.042), 1.0% (RHC, 0.925–0.934) and 0.5% (Combined, 0.973–0.978), and equal to the one-bin totals (1.042, 0.925, 0.975), whereas the Wiener-SVD integrals spread by 30%, 27% and 23%. The observable-to-observable spread in the Wiener-SVD integrated cross sections (Table 23 of the analysis note) is therefore the A_C smearing rather than a physical or acceptance effect.

Table 18: Wiener-SVD vs. D'Agostini (iterative Bayesian, 4 iterations) integrated cross section, averaged over the five inclusive observables (p_μ, three-region p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$; 10^{-38} cm^2/Ar), evaluated on the released pseudo-data (`xsec_analyzer/dagostini_050.sh`). D'Agostini carries no A_C smearing, so it recovers a 27–30% higher total, and it is flat across observables (spread 1.0% FHC, 1.1% RHC, 0.5% combined) where the Wiener-SVD integrals spread by 23–30%, so the observable-to-observable variation of σ_{int} in Table 23 of the analysis note is the A_C smearing rather than a physical or normalisation effect. The last column is the change of the D'Agostini integral between 1 and 4 iterations.

Config	$\langle\sigma\rangle_{\text{WSVD}}$	$\langle\sigma\rangle_{\text{DAg}}$	DAg/WSVD	DAg 1→4 iter
FHC	0.798	1.039	1.30	$\leq 0.3\%$
RHC	0.712	0.927	1.30	$\leq 0.3\%$
Combined	0.768	0.975	1.27	$\leq 0.1\%$

13 Identity closure: unfolding the prediction with itself [SHARED]

The pseudo-data closure tests reported in the analysis notes compare an unfolded Poisson throw against the truth it was thrown from. They therefore test the whole chain at once, and a modest χ^2 can hide a genuine inconsistency inside the unfolding itself, because the statistical fluctuation of the throw is the dominant term.

The identity test removes that fluctuation. Wiener–SVD returns $\hat{x} = Ur$ for a measured reco vector r, and defines the additional-smearing matrix as $A_C = UR$, where R is the response matrix. Feeding an exact prediction $r = Rt$ back through the chain must therefore return

$$\hat{x} = URt = A_C\, t \tag{1}$$

identically, for any true vector t. Rather than pick one t, we test the matrix identity directly:

$$\left\| UR - A_C \right\|_\infty \stackrel{!}{=} 0. \tag{2}$$

This is stronger than a single-vector test and independent of the data. It is sensitive to the failure mode a pseudo-data test cannot see: the unfolding matrix and the additional-smearing matrix being built from different responses, or from responses evaluated in different universes, which would bias every extracted cross section by a fixed factor while leaving the closure χ^2 untouched.

The check is implemented inside `UnfolderNuMI` and runs on every extraction, printing a `[IDENTITY]` line. All 59 released extractions pass it: five inclusive and twelve proton-tagged observables, each in three beam configurations, the two proton angles in three configurations and the two combined two-dimensional results. The worst deviation over the 59 extractions is 5.7×10^{-5} relative, on FHC $\cos\theta_\mu$; the $\cos\theta_\mu$ extractions of the two families are the only ones above 6×10^{-6}. These are the rounding errors of the singular-value decomposition and the matrix products that follow it, not physical effects: they remain more than two orders of magnitude below the smallest systematic uncertainty in the analysis.

The deviations are larger than the 10^{-10} level obtained with regularisation matrices that are not width-aware, and the reason is arithmetic rather than physical. The width-aware operator carries factors of $1/w_j$ that span decades across a binning with a width ratio of up to 56:1, so the matrix products accumulate more floating-point error. $\cos\theta_\mu$, whose eight bins have the largest width ratio in the analysis, is the worst case. The identity itself holds exactly; only the arithmetic that verifies it is noisier. The unfolding construction is therefore exactly self-consistent, and every deviation seen in the pseudo-data closure tests is attributable to the statistical throw and to the response model, not to the unfolding implementation.

14 Total cross section and systematic breakdown [INCLUSIVE]

The measurement is the one-bin extraction of Sec. 14.6 of the analysis note on the Poisson-thrown pseudo-data (one true bin over the full $\cos\theta_\mu$ range, no overflow, Wiener filter off, $A_C \equiv 1$; `data_release/total_xsec.tsv`), with its full universe-propagated uncertainty; at unblinding it becomes the real beam-on extraction. It agrees with all four generators within 0.8σ: standalone GENIE sits 0.6–0.8σ low, while GiBUU, NEUT and NuWro lie within 0.7σ.

Table 19: Total flux-averaged cross section from the one-bin extraction (Sec. 14.6 of the analysis note; 10^{-38} cm^2/Ar), compared with the flux-averaged generator predictions. The measurement uncertainty is the official one-bin covariance (flux, cross section, re-interaction, detector, POT, targets and statistics; the MCS momentum-scale term is zero for the total, which uses no reconstructed muon momentum; the beamline-geometry, beam-off trigger-ratio and Run-2 stand-in terms lie outside it, Sec. 12.4 of the analysis note). `data_release/total_xsec.tsv`. Values in parentheses give each generator's deviation from the measurement in units of the measurement uncertainty. The generator predictions are integrated over the full binning with open upper bins (a closed upper edge on p_μ or p_π would discard 4–8% of each prediction).

Config	Measurement	GENIE	GiBUU	NEUT	NuWro
FHC	1.04 ± 0.37	$0.82\,(+0.6\sigma)$	$1.13\,(-0.2\sigma)$	$1.32\,(-0.7\sigma)$	$1.25\,(-0.6\sigma)$
RHC	0.92 ± 0.32	$0.67\,(+0.8\sigma)$	$0.94\,(-0.1\sigma)$	$1.10\,(-0.6\sigma)$	$1.04\,(-0.4\sigma)$
Combined	0.97 ± 0.34	$0.73\,(+0.7\sigma)$	$1.02\,(-0.1\sigma)$	$1.20\,(-0.7\sigma)$	$1.13\,(-0.5\sigma)$

The systematic breakdown by source (bin-averaged fractional uncertainty in cross-section space) is given for all three configurations in Tables 20–22. The flux (PPFX) term dominates throughout, followed by the detector variations; the data-statistical term reflects the pseudo-data

POT of each configuration. The largest detector entry is that of FHC p_π, 24.9%; elsewhere the detector term is 12.1–19.3%. The tables use the official covariance: the MCS momentum-scale row is the data-side term, computed outside the framework and non-zero only for p_μ (Sec. 12.3 of the analysis note).

Table 20: FHC ({Run 1,2,4,5}) systematic breakdown by source, bin-averaged fractional uncertainty (%), from the released systematics dumps (`logs/systdump`); "Prediction total" excludes the data statistics. The MCS momentum-scale row is the data-side term that the official covariance adds to the framework covariance.

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Prediction total	**37.0**	**49.4**	**36.7**	**35.8**	**33.5**
Cross section (GENIE)	11.0	15.8	11.2	11.3	10.1
Flux (PPFX)	28.6	36.5	28.0	27.1	28.3
Detector	16.1	24.9	18.8	17.6	13.5
Reinteraction	4.1	12.8	3.5	5.1	3.8
MCS momentum scale	10.1	0.0	0.0	0.0	0.0
MC stat	3.4	5.2	5.4	4.4	2.5
EXT stat	0.7	1.8	1.6	1.2	0.8
Data stat	10.9	16.4	17.0	14.1	7.9
POT + targets	3.6	4.7	3.6	3.4	3.5
Total (incl. data stat)	**38.6**	**52.1**	**40.7**	**38.6**	**34.5**

Table 21: RHC ({Run 1,2,3,4a,4b,4c}) systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 20.

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Prediction total	**38.8**	**37.5**	**34.0**	**33.7**	**31.8**
Cross section (GENIE)	11.3	13.9	9.6	10.5	11.0
Flux (PPFX)	27.9	29.2	25.9	25.9	26.2
Detector	19.3	14.4	17.4	16.9	12.5
Reinteraction	4.0	10.5	4.0	5.0	3.7
MCS momentum scale	13.4	0.0	0.0	0.0	0.0
MC stat	3.0	3.8	4.6	3.7	2.5
EXT stat	1.0	1.1	1.0	1.0	0.7
Data stat	10.3	13.0	15.9	12.7	8.6
POT + targets	3.8	4.1	3.6	3.6	3.6
Total (incl. data stat)	**40.3**	**39.8**	**37.7**	**36.2**	**33.0**

Table 22: Combined-configuration (FHC+RHC) systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 20.

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Prediction total	**35.6**	**41.5**	**32.1**	**31.1**	**31.7**
Cross section (GENIE)	10.7	14.8	9.5	10.4	10.7
Flux (PPFX)	26.3	31.8	24.9	24.1	26.6
Detector	17.6	17.4	16.1	13.9	12.1
Reinteraction	3.9	12.3	3.9	4.8	3.6
MCS momentum scale	9.9	0.0	0.0	0.0	0.0
MC stat	2.7	3.5	3.5	3.3	2.1
EXT stat	0.6	1.1	0.9	0.9	0.6
Data stat	9.1	11.6	11.5	10.8	7.1
POT + targets	3.5	4.3	3.4	3.2	3.5
Total (incl. data stat)	**36.8**	**43.1**	**34.3**	**33.0**	**32.5**

15 Presentation of the results [INCLUSIVE]

15.1 The three-region p_π result

Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/c: a pion of true momentum 0.95 GeV/c reconstructs at almost the same value as

one of 0.55 GeV/c. The three regions have migration diagonals of 91%, 51% and 52% (column-normalised M, Sec. 11.2 of the analysis note) and meet the 0.50 criterion, but the upper region integrates over a very wide physical interval and is not a differential point in the usual sense. The true upper region is open-ended, $p_\pi > 0.26\,\mathrm{GeV}/c$ (the true-bin definition places no upper edge on it); the 1 GeV/c that appears in the data release and in Appendix A of the analysis note is only the finite boundary adopted to assign the region a width for plotting, not a kinematic limit. The released $d\sigma/dp_\pi$ values are therefore bin-averaged densities over very different intervals (0.030, 0.055 and 0.74 GeV/c), and the panel draws the three at equal visual width, flagged on the axis, so the plotted area does not represent cross section. The result is stated as the three bin-integrated partial cross sections (Table 23); the density representation derives from them and depends on the conventional plotting width of the open region. These are the partial cross sections, $\sigma_i = (d\sigma/dp_\pi)_i\,\Delta p_{\pi,i}$, which do not depend on the choice of plotting boundary for the two lower regions and depend on it for the upper one only through the physical content above 1 GeV/c, which the open true bin already includes. For the open upper region the released density and the conventional width of 0.74 GeV/c form a matched representation of the extracted bin integral: a user should compare bin-integrated predictions after applying A_C, not reinterpret the density over a different finite upper boundary, since changing the boundary while holding the released density fixed would change the product:

Table 23: Three-region p_π result as bin-integrated partial cross sections (10^{-38} cm^2/Ar), computed from the released `curves_incl_*_ppi3bin.tsv` as density × physical width; the uncertainty is the total uncertainty from the released covariance (the same fractional uncertainties as Tables 20 of the analysis note–22). All three regions are A_C-smeared like every other unfolded quantity, so a prediction must be smeared by the released A_C before comparison.

Region	FHC	RHC	Combined
$\sigma(0.175 < p_\pi < 0.205\,\mathrm{GeV}/c)$	0.056 ± 0.035	0.066 ± 0.028	0.062 ± 0.030
$\sigma(0.205 < p_\pi < 0.26\,\mathrm{GeV}/c)$	0.096 ± 0.059	0.112 ± 0.047	0.105 ± 0.050
$\sigma(p_\pi > 0.26\,\mathrm{GeV}/c)$	0.691 ± 0.229	0.565 ± 0.199	0.611 ± 0.205

The saturation is a physical information limit rather than a shortcoming of the range estimator, as the survey of pion momentum estimators shows (§6.3).

15.2 Interpretation of the closure χ^2 values

The closure χ^2 quoted for each extraction compares the unfolded result with the pseudo-data truth using the complete configured systematic covariance, and the resulting p-values cluster near unity (median 0.97 across all fifty-one extractions; thirteen lie below 0.9 and the lowest is 0.12). This is not a sign of excellent agreement with the model. For a self-closure with correctly-sized uncertainties the p-values would be roughly uniform on $[0, 1]$, and they are not.

The reason is that the measurement and the truth it is being compared against share their systematics: both come from the same generator, the same flux and the same exposure, so the flux, cross-section and exposure terms are common-mode and largely cancel in the difference while still inflating the covariance used to judge it. Recomputing the same comparison with the statistical covariance alone raises χ^2/ndf from 0.02–0.2 to 0.5–0.8, much nearer the expected value. The table below was evaluated on a different pseudo-data throw from the released one and illustrates the effect; the released closure values are those of Tables 20 of the analysis note–22.

Extraction	full	prediction-only	statistics-only
inclusive FHC p_μ	0.14/7	0.22/7	1.01/7
inclusive comb p_μ	1.43/7	2.87/7	4.71/7
proton-tagged FHC p_μ	1.01/7	2.20/7	3.02/7
proton-tagged comb $W_{\pi p}$	0.48/6	0.77/6	4.88/6

The closure χ^2 is therefore a check that the extraction does not disagree with its input beyond the quoted band, which it passes, and not a measure of how well the band is sized. A test of the latter needs either a covariance restricted to the terms that differ between measurement and reference, or an ensemble of throws. The restricted covariance is not used for the released closure tables, whose p-values should be read with that limitation in mind; the sizing is tested instead with the ensembles of Sec. 13.2 of the analysis note and §17.

16 A_C row sums of every extraction [SHARED]

Table 24 gives the row sums of the released additional-smearing matrices of every inclusive and proton-tagged extraction. No matrix has a negative row.

Table 24: Range of the A_C row sums of every released extraction, from the result files.

Observable	FHC	RHC	combined
Inclusive			
p_μ	0.48–1.03	0.39–0.75	0.49–0.81
p_π (three regions)	0.42–0.83	0.40–0.94	0.38–0.88
$\cos\theta_\mu$	0.50–0.73	0.44–0.79	0.51–0.71
$\cos\theta_\pi$	0.66–0.76	0.71–0.81	0.56–0.77
$\theta_{\mu\pi}$	0.54–1.14	0.59–1.13	0.65–1.22
Proton-tagged			
p_μ	0.38–1.02	0.35–0.84	0.47–0.74
p_π (two regions)	0.38–0.77	0.39–0.51	0.48–0.52
$\cos\theta_\mu$	0.51–1.17	0.44–0.95	0.49–0.88
$\cos\theta_\pi$	0.68–0.78	0.65–0.77	0.71–0.80
$\theta_{\mu\pi}$	0.36–0.85	0.35–0.77	0.35–0.76
p_p	0.64–0.80	0.71–0.88	0.69–0.86
$W_{\pi p}$	0.46–1.21	0.24–1.24	0.46–1.20
W_{had}	0.81–0.89	0.80–0.87	0.83–0.87
δp_T	0.48–0.86	0.45–0.86	0.59–0.89
$\delta\phi_T$	0.26–0.71	0.30–0.74	0.49–0.79
$\delta\alpha_T$	0.62–0.93	0.61–0.92	0.64–0.90
p_n	0.65–0.86	0.65–0.86	0.58–0.86

Three features bear on the interpretation of the results.

The suppression is large: inclusive p_μ has a row sum as low as 0.39 in RHC, and the proton-tagged $\delta\phi_T$ one of 0.26, a strong suppression of that output mode for a flat input. It also differs strongly between observables, which is why the integrated cross section obtained by summing a differential result varies from observable to observable while the underlying physical total cannot. The D'Agostini cross-check, whose integral is flat across observables, establishes that this spread is the smearing rather than a physical or acceptance effect.

In $W_{\pi p}$ the top row carries negative off-diagonal terms in all three configurations (summing to -0.14 FHC, -0.05 combined, -0.19 RHC against diagonals of 0.69, 0.51 and 0.42); its row sum is lowest in RHC (0.24), because its diagonal element is the smallest of the three. The first $\cos\theta_\mu$ bin of the proton-tagged RHC sample has a positive row sum with the eight adopted bins.

This is therefore not an RHC-specific defect but the regularisation pushing back against the least-determined bin of the observable: for $W_{\pi p}$, the same top bin that is only 2% diagonal in the six-bin response and is the reason no differential $W_{\pi p}$ result is reported. A negative element is not forbidden, since A_C is a regularisation operator rather than a probability matrix, but a prediction smeared through one is being told that raising the truth in that bin lowers the measured content, which is a statement about the regularisation and not about the detector. That row should be used with corresponding caution.

17 Ensembles: reference construction and full-covariance pulls

[SHARED]

Ensemble coverage and the size of the uncertainties. The closure test above uses a single throw. It can show that the extraction reproduces its input, but a single residual has no width, so it cannot say whether the quoted uncertainty is correctly sized. To test that, one hundred statistically independent Poisson realisations of the data of each configuration (six extractions: p_μ and $\cos\theta_\mu$ in FHC, $\cos\theta_\mu$ in RHC, the three-region p_π in the combined configuration, the FHC and RHC one-bin totals; 100 members each) were each put through the whole chain (thrown, universes rebuilt from that throw, unfolded), and the per-bin pull

$$p_i \;=\; \frac{\hat{x}_i - (A_C\, x_{\mathrm{CV}})_i}{\sigma_i}$$

formed against the fixed central-value truth smeared by that member's own A_C. (The per-throw pseudo-data truth is not a valid reference: it follows the Poisson draw, so numerator and reference fluctuate together and the width comes out too small.)

The throws follow the weight rule of Eq. 4 of the analysis note, and the extractions are chosen to bracket the conditioning and the configurations: the well-behaved p_μ; $\cos\theta_\mu$, whose A_C row sums fall as low as 0.50 in FHC (0.44 in RHC) so that the regularisation removes up to 50% (56%) of a bin's normalisation; the three-region p_π of the combined configuration; and the one-bin totals, which have no regularisation at all. The four differential ensembles use the released binnings (members built from staged configurations identical to the released ones). If the uncertainties were mis-sized anywhere, it would show in $\cos\theta_\mu$; if the extraction carried a bias independent of the regularisation, it would show in the one-bin totals.

Reference covariance	pull mean	pull width	68% / 95% coverage
statistical only			
$d\sigma/dp_\mu$, FHC	−0.10	1.03	67.3% / 94.5%
$d\sigma/d\cos\theta_\mu$, FHC	−0.05	1.00	69.8% / 95.0%
$d\sigma/d\cos\theta_\mu$, RHC	−0.05	1.04	63.5% / 93.5%
$d\sigma/dp_\pi$ (three regions), combined	−0.08	0.94	72.3% / 95.3%
one-bin total, FHC	−0.04	0.92	71.0% / 96.0%
one-bin total, RHC	−0.16	1.01	66.0% / 95.0%
full			
$d\sigma/dp_\mu$, FHC	−0.02	0.31	99.7% / 100.0%
$d\sigma/d\cos\theta_\mu$, FHC	−0.02	0.42	97.6% / 100.0%
$d\sigma/d\cos\theta_\mu$, RHC	−0.01	0.44	97.5% / 100.0%
$d\sigma/dp_\pi$ (three regions), combined	−0.02	0.25	100.0% / 100.0%
one-bin total, FHC	−0.00	0.11	100.0% / 100.0%
one-bin total, RHC	−0.02	0.12	100.0% / 100.0%
expected	0	1	68% / 95%

Against the statistical covariance all six widths are consistent with unity (0.92–1.04, each determined to about ±7%, $1/\sqrt{2\times 100}$, more for the multi-bin extractions, whose bins are correlated), and the interval coverage is correct within the same statistics. The statistical uncertainties are therefore correctly sized, and the strongly suppressed observable behaves no differently from the well-conditioned one. Of the two, the $\cos\theta_\mu$ result is the more informative: p_μ was expected to pass, whereas it was not known in advance whether the severe A_C suppression in $\cos\theta_\mu$ would distort its uncertainties.

Against the full framework covariance, without the MCS term, the pull width is 0.25–0.44 for the differential results and 0.11–0.12 for the totals, and coverage is nearly 100%: the quoted uncertainty is two to four times the throw-to-throw spread for the differential results and

about nine times for the totals, because the ensemble varies only the data statistics while the denominator carries flux, detector and cross-section terms that are identical in every member. This is the same effect as the closure p-values clustered near unity, measured here directly.

The ensemble means sit below the reference in all six cases, by 0.1–0.9% (0.3–1.5σ on the mean; largest in significance for the RHC total, -0.60% at 1.5σ, and in size for FHC $\cos\theta_\mu$, -0.89% at 1.1σ), with pull means of -0.04 to -0.10 and -0.16 for the RHC total (Table above; `data_release/ensemble_2026-10-09.tsv`). The extractions of one configuration share the same throws and are correlated, so this is a few measurements of a small negative offset, each consistent with zero at the present ensemble size. The throw itself is unbiased: the mean selected count over the 100 members of the FHC ensembles is 1497.7 ± 3.6 against the neutrino-MC expectation of 1492.2 (ratio 1.004), so whatever offset remains is in the extraction, not in the pseudo-data. The offset is at most two standard deviations of the ensemble mean and is largest in FHC $\cos\theta_\mu$, where the regularisation is strong, but the one-bin totals, which have no regularisation, also sit low, so the excess in FHC $\cos\theta_\mu$ over the common $\approx 0.5\%$ is the only part that can be attributed to A_C. No correction is applied and no uncertainty is assigned; a larger ensemble would settle whether the common offset is real.

A truncated read of the pseudo-data occurs when two ensembles sharing a throw number write the same throw file at the same time and a member opens it mid-write, seeing part of the tree. In an earlier round, ten of the 600 members were first built from such a read. Four RHC-total members (Run 2 read at 47–72%) and one FHC $\cos\theta_\mu$ member (43%) were affected materially; they widened the RHC-total pull width to 1.28. Every member was checked for this (the number of events univmake read against the entries on disk; 2791 of 2800 inputs agree exactly, and the five remaining one-entry differences are one out-of-range truth event in the one-bin binning, which does not enter the reco-level data), and the affected members were rebuilt. Each throw is created once and never replaced (`macros/throw_guard.h`; change log). The seventeen ensembles of version 1.7 were built this way, and the entry-count scan of their 9200 inputs finds no truncated read; the 71 inputs short by one or two entries are the boundary case of an event at the edge of the true binning, $\cos\theta_\mu = 1.0$ exactly or just outside $[-1, 1]$ in the one-bin binning (change log).

The study covers the inclusive sample in all three configurations. The two-dimensional results and the proton angles have their own 100-member ensembles: the inclusive $\cos\theta_\pi \times \cos\theta_\mu$ (Sec. 14.7 of the analysis note), and on the proton-tagged sample θ_p and $\theta_{\pi p}$ in each configuration and $\theta_p \times \delta p_T$, with statistical pull widths of 0.92–1.04 (`data_release/ensemble_v1_2026-10-09.tsv`). The pulls of a two-dimensional result are read from its all-slice closure file, which holds every analysis bin in the order of the covariance. The proton-tagged family, where the statistics are sparser, has in addition three ensembles of validation-only products (W_{had} in its two regions and the two-bin δp_T in FHC, the two-bin p_n combined): statistical pull widths 0.93–1.09, 68% coverage 64–73% and integral offsets within 0.9% (Sec. 5.2 of the proton-tagged note, `data_release/ensemble_1p_2026-10-09.tsv`).

18 Consistency between configurations and with the sibling ν_e measurement [INCLUSIVE]

18.1 Consistency with the sibling ν_e CC 1π measurement

This analysis is one of a pair of MicroBooNE NuMI CC single-charged-pion measurements; the companion ν_e CC 1π analysis shares the same cross-section-extraction framework, beam and reconstruction. A cross-check against its internal note confirms consistency on the essential conventions and identifies the differences that a combined NuMI CCπ programme should reconcile.

Consistent by construction. Both use the Wiener–SVD extraction framework; the updated NuMI flux with the fixed geometry bug and Geant v4.10.4; the same signal topology (exactly one

$\pi^{\pm}$, zero π^0 or heavier mesons, any number of protons, charged current), and the same systematic suite, PPFX flux hadronic universes (the 12 beamline-geometry unisims the ν_e analysis carries are evaluated but not in this analysis's official covariance, Sec. 12.4 of the analysis note), the 600-universe GENIE cross-section multisim with unisims, the Geant4 reinteraction reweighting, the standard detector-variation set, and 2% / 1% POT / target normalisation uncertainties. The FHC beam-on data POT per run agrees exactly (Run 2 1.268, Run 4 2.075×10^{20}).

Differences reconciled or flagged. (i) **POT accounting:** the ν_e note quotes Run 1 FHC $= 3.283 \times 10^{20}$ POT (9 846 635 triggers), the figure that includes the open-trigger stream. The beam-on sample processed here is the non-open-trigger dataset, 2.192×10^{20} POT with 5 748 692 triggers, confirmed against the beam bookkeeping; the total FHC exposure is 7.766×10^{20} over Runs 1, 2, 4 and 5. (ii) **Run 4:** the `Run4` overlay is identical to `Run4c`+`Run4d` combined (same 859,368 events and 2.834×10^{21} POT); only one may be used to avoid double counting. (iii) **Pion threshold:** ν_e uses $\mathrm{KE}_\pi > 40$ MeV ($p_\pi > 0.113\,\mathrm{GeV}/c$) versus our $p_\pi > 0.175\,\mathrm{GeV}/c$ (Table 8 of the analysis note). (iv) **Opening-angle cut:** ν_e excludes $\theta_{e\pi} > 170°$ (a shower-specific back-to-back misreconstruction), whereas our muon-track topology uses $\theta_{\mu\pi} < 2.6$ rad (149°). (v) **Regularisation:** the ν_e analysis smooths on the first derivative, this analysis on the second (§11). (vi) **Scope:** both analyses cover Runs 1–5 in FHC and RHC; the present measurement reports FHC, RHC and the combined configuration. (vii) **Dirt:** the ν_e analysis assigns a 100% dirt normalisation uncertainty; this analysis assigns none. The dirt is 6.6, 9.5 and 16.0 selected events (FHC, RHC, combined), 0.8–0.9% of the selected signal, so a 100% variation moves the one-bin totals by at most 0.9%, negligible against their 34–36% uncertainty.

18.2 The combined configuration and its inputs

The integrated cross section extracted from a given configuration is not a configuration-independent number: it depends on the additional-smearing matrix A_C, which differs per configuration and per observable, so comparisons across configurations are made observable by observable and against the pseudo-data truth rather than between measurements. Two tests compare the combined configuration with its inputs.

The combined truth against the exposure-weighted average of its inputs. A configuration spanning both horn currents should predict a cross section between those of its two inputs, equal to their average weighted by the fluence of each mode (0.425 FHC, 0.575 RHC). The realised truths of the one-bin totals (Table 27 of the analysis note) give a ratio of the combined truth to that average of 1.000 for the inclusive family and 1.001 for the proton-tagged family. Observable by observable, the A_C-smeared truth integrals give 1.003 (0.986–1.019) and 1.035 (0.989–1.229), the spread being the configuration dependence of A_C rather than of the truth. The test is sensitive to the treatment of the detector variations: the two per-mode detector-variation sets of the combined configuration must be pooled as one sample scaled once to the data exposure, since scaling each set to the full exposure and summing would double the detector covariance, over-regularise the combined extraction and push its truth ratio to 0.91 with a worst observable at 0.77. With the pooled treatment the combined detector term lies between or close to those of its inputs (analysis note, Tables 20–22: 17.6% against 16.1% in FHC and 19.3% in RHC for p_μ).

The RHC deficit is in the input, not the extraction. The one-bin totals give an RHC/FHC ratio of the injected tune of 0.901 (inclusive) and 0.878 (proton-tagged), and the extracted one-bin ratios are 0.901 and 0.873 (Table 27 of the analysis note). Normalisation is not implicated: the exposures sum exactly ($7.766 \times 10^{20} + 1.1082 \times 10^{21} = 1.8848 \times 10^{21}$ POT) and the per-configuration flux values reproduce the POT-weighted combination to 0.01%. The deficit is therefore a property of the $\bar{\nu}_\mu$-dominant sample, and the number to be compared with data at unblinding is the tune ratio.

External reproduction of the released curves. The transformation $(d\sigma/dx)_i = w_i^{-1} \sum_j (A_C)_{ij}\, p_j$ applied to the released truth-level predictions reproduces every released generator curve to a worst relative per-bin deviation of 4×10^{-8}, the precision of the text files (96 curves). This check is part of the release procedure because an internal closure cannot make it: the unfolded data and the truth can both be right while the model comparison is wrong, and only an external user's arithmetic exposes it.

19 Control regions: definitions and roles [INCLUSIVE]

The four reconstruction-level control regions of the inclusive CC $1\mu 1\pi Xp$ selection are CC0π, multi-π, π^0 and large muon–pion opening angle (called "cosmic" below for brevity: it is cosmic-enriched, 26–33% beam-off, not cosmic-pure); Sec. 10.1 of the analysis note gives their definitions. Throughout, "beam-off", "EXT" and "cosmic data" name the same data-driven component, the beam-off trigger sample scaled by trigger counts; "signal contamination" means the inclusive signal unless the proton-tagged selection is named. Each region is anchored by the inversion of one nominal signal requirement; requirements that become undefined once that inversion is made, or that would sculpt the targeted control sample, are dropped as listed in Table 25. Each region is disjoint from the signal region at reconstruction level (demonstrated event by event, §20.1); their expected signal contaminations are 8–24%. The selections, the run-dependent normalisation, the beam-off scaling, the displayed observables and binning, the covariance construction, the goodness-of-fit tests and the responses to a discrepancy are fixed in §20.5 before beam-on data in the regions are examined. The data are compared with the complete prediction (neutrino MC + beam-off + dirt) separately in FHC and RHC, with the combined configuration as a cross-check. The proton-tagged control regions are discussed in §20.6.

Role of each region. The transfer factors of §20.3 set the physics role of each region.

- **CC**0π is the high-statistics validation of the dominant neutrino background class: CC0π events with a proton or other track taken as the pion are 260 of the 620 FHC neutrino-background events (42%). The region has 72% target purity (64% of the full stack) and 0.6% raw statistical precision. It validates the normalisation and kinematics of the underlying CC0π population, not the proton-to-pion misidentification probability that transfers that population into the signal region: the transfer carries a 24–28% uncertainty dominated by the detector variations. It is therefore a validation test rather than a per-cent-level constraint, and it is conditionally eligible for a fitted normalisation correction under §20.5. The direct test of the misidentification probability would be an inverted pion-PID region, which has not been defined.
- π^0 tests the shower veto and the CC+NC π^0 background. Its target purity is 50.1% (FHC) and 51.0% (RHC) of the neutrino MC, its transfer factor 0.02 with a 18–22% uncertainty set by the Monte-Carlo-statistical floor of the detector-variation samples. The region is diagnostic only; a fit of the CC+NC π^0 normalisation is considered only after the pre-fit result is recorded, together with the fit model, the leverage of the fixed components, the transfer uncertainty and its floor, and the effect on the signal-region prediction.
- **Multi-**π is diagnostic only. Its transfer factor is $\simeq 2.3$ (the signal region holds more of the target class than the region does) and its target purity 45–46%; it is not used to fit or constrain the signal-region normalisation, but it exposes gross modelling failures and shape discrepancies.
- **Cosmic** validates the beam-off normalisation and shape to $\sim 50\%$ once the prediction uncertainties of its neutrino component are included (17% from data statistics alone). This precision is sufficient because beam-off is 1.4% of the selected signal-region prediction: a

50% uncertainty on it is an effect of order 1% on the selected sample. It is not eligible for a fitted correction.

About 30% of the FHC neutrino background has no region of its own: NC and ν_eCC without a π^0, out-of-fiducial-volume events and CC1$\pi^\pm$ outside the measured phase space (§20.3). These components keep their model uncertainties unchanged; the CC0π and π^0 distributions carry indirect sensitivity to the first two through the shared reconstruction, and each is small enough individually that no dedicated region is justified at this exposure.

Sensitivity of the test and its blind direction. The pass/fail statistic is a global normalisation χ^2 over the eight (region, mode) totals with the full pre-fit covariance (§20.5). It detects a discrepancy confined to one region beyond 12–13% of the CC0π or π^0 total and beyond 45–52% of the multi-π or cosmic total, an FHC/RHC inconsistency, and, through the normalisation-marginalised shape tests, a shape difference. A common normalisation shift of all regions fails it only beyond 50%, because the correlated flux, cross-section and detector terms absorb it. That blind direction is acceptable because it is the direction the signal-region measurement already carries as its flux and cross-section uncertainty: the control regions test the background model region by region, not the overall normalisation.

Scope of the pseudo-data studies. The control-region machinery has been exercised on Poisson-thrown central-value pseudo-data. Those studies validate the selection logic, the per-run exposure scaling, the central-value weighting and, in the full-stack variant of §20.2, the exact bookkeeping path that beam-on data will follow. They do not validate the background model: the pseudo-data are drawn from the same simulation that supplies the prediction, so agreement is guaranteed up to statistics. The background model is the largest untested assumption in the analysis, and the control regions exist to test it. The control-region comparison is made at reconstruction level, so it does not depend on the unfolding or on a closure test with an independent generator.

Table 25: Control-region definitions in terms of the selection stages. The common preselection *pre* is software trigger, reconstructed vertex in the fiducial volume, topological score > 0.67 and a qualifying muon candidate (Cuts 1–4 of Sec. 8 of the analysis note); *shower* is the shower veto (Cut 8), *final* the non-proton multiplicity cut (Cut 10), *contained/gap* the pion containment and three-plane requirements (Cuts 5–7), *opening* the $\theta_{\mu\pi} < 2.6$ rad cut (Cut 9), and n_π the number of qualifying contained pion candidates. The signal region is *pre*, *contained*, *gap*, *shower*, *opening* and *final* with $n_\pi = 1$. "Dropped" lists nominal requirements not applied in the region; "target" the signal-region background the region is meant to constrain. Each region is stored as a flag alongside the signal-region flag when the events are processed.

Region	Definition (branch level)	Dropped vs. signal	Target background
CC0π	*pre*, *shower*, *final*; $n_\pi = 0$	contained, gap, opening (no pion to define them)	CC0π events with a proton or other track taken as the pion; pion absorption feed-in
multi-π	*pre*, *final*; $n_\pi \geq 2$	shower, contained/-gap of the extra pion, opening	multi-pion events with one pion lost
π^0	*pre*, shower veto failed; $n_{\text{shower}} \geq 1$	contained, gap, opening, final	CC/NC π^0 with both photons missed; shower-veto efficiency
cosmic	*pre*, *contained*, *gap*, *shower*, *final*; opening-angle cut inverted	opening only	residual beam-off (cosmic) in the signal region; its normalisation

The π^0 region is the furthest from a single inversion: besides the pion requirements, which have

no meaning once no pion candidate is demanded, it also drops the final multiplicity cut. The cut is dropped to avoid a plausible topology-dependent sculpting (photon showers reconstructed as additional track-like primaries). The comparison of §20.4 finds no material change in target purity, transfer uncertainty or the frozen leading-shower-energy shape. The no-cut definition is retained because it provides 11–14% more events, not because the study demonstrated a material bias from the cut.

Expected yields and precision. Table 13 of the analysis note gives the expected composition per region and beam. The raw statistical precision of the data samples is 0.6% (CC0π), 0.8% (π^0) and 6% (multi-π) in the combined configuration, and the prediction in each region carries 21–35% flux, cross-section, re-interaction and detector uncertainties (§20.3). The regions are blind to the signal at this precision (§20.5, item 4).

20 Control-region machinery [INCLUSIVE]

20.1 Region overlaps

The four regions are disjoint from the signal region by construction: CC0π and multi-π require $n_\pi = 0$ and $n_\pi \geq 2$ against the signal region's $n_\pi = 1$, the π^0 region requires the shower veto to fail and the cosmic region the opening-angle cut to fail. On the files carrying the region flags, the number of events with both the signal-region flag and any control-region flag is exactly zero in the simulation and in the pseudo-data of every configuration. Among themselves the regions overlap in one place: multi-π does not apply the shower veto, so 38% (FHC) and 42% (RHC) of its simulated events also belong to the π^0 region. A simultaneous fit of the two would have to account for that; the per-region comparisons of §20.5 do not.

20.2 Full-stack pseudo-data test

To exercise the path that beam-on data will take, the regions were compared with pseudo-data made of the neutrino-simulation throw plus an independent Poisson throw of the beam-off and dirt expectations, against the complete prediction. The data-to-prediction ratios are 0.94–1.05 in FHC, 1.00–1.14 in RHC and 1.00–1.06 combined, each within 1.9 times its statistical uncertainty (0.6–9%). This tests the bookkeeping, not the physics. Figure 7 shows the frozen FHC distributions with these pseudo-data in place of the beam-on points; the RHC distributions are drawn from the same list.

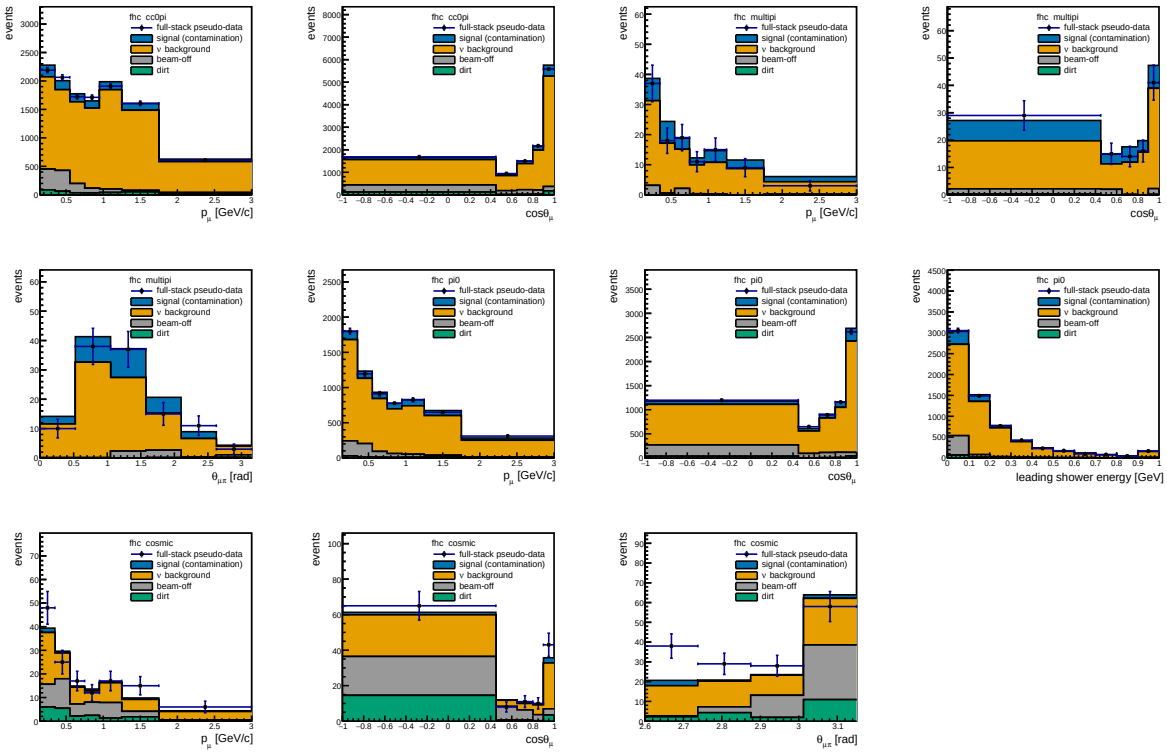


Figure 7: FHC control-region distributions at the frozen binning: by row, CC0π, multi-π, π^0 and cosmic; p_μ, $\cos\theta_\mu$ and, where defined, the muon–pion opening angle or the leading shower energy. Stack: signal contamination, neutrino background, beam-off (scaled per run) and dirt; points: full-stack pseudo-data. Overflow is in the last bin.

20.3 Transfer factors and control-region uncertainties

A control-region discrepancy can be propagated into the signal region only through the transfer factor $T_{b,c} = N_b^{\mathrm{SR}}/N_b^{\mathrm{CR}_c}$ of each background class b, and a region constrains a class only where T is small and stable against the model variations (Table 26).

Table 26: Composition of the signal region (SR) and of each control region by background class (central-value neutrino simulation scaled per run; beam-off and dirt excluded) and the transfer factors $T = N^{\mathrm{SR}}/N^{\mathrm{CR}}$. CC0$\pi$ constrains the 0π class ($T \simeq 0.03$) and π^0 the π^0 class ($T \simeq 0.02$; the adopted target class CC+NC π^0, last row of each block, is 50–51% of the region's neutrino simulation); the multi-π region holds fewer multi-pion events than the signal region ($T \simeq 2.3$); the cosmic region constrains only the beam-off component (42 beam-off events against 24 in the signal region, $T = 0.56$, FHC).

Class	SR	CC0π	multi-π	π^0	cosmic	$T_{\mathrm{CC0}\pi}$	T_{multi}	T_{π^0}	T_{cosmic}
FHC									
signal (contamination)	872	943	29.2	582	5.4	0.93	30	1.50	162
ν_μCC 0π (proton or other track as pion)	260	7 642	2.6	887	20.2	0.03	98	0.29	13
ν_μCC 1$\pi^\pm$ outside the measured phase space	49.6	550	2.7	318	13.5	0.09	19	0.16	3.68
ν_μCC with a π^0	52.7	357	19.9	2 702	2.1	0.15	2.64	0.02	25
ν_μCC $\geq 2\pi^\pm$, no π^0	131	189	55.6	582	3.6	0.70	2.36	0.23	37
out of fiducial volume	44.0	415	3.1	222	16.7	0.11	14	0.20	2.63
NC and ν_eCC	82.5	477	7.0	555	6.0	0.17	12	0.15	14
total ν-MC	1 492	10 571	120	5 847	67.4				
CC+NC with a π^0 (adopted π^0 target; overlaps two rows above)	58.6	389	21.4	2 931	2.7	0.15	2.74	0.020	22
RHC									
signal (contamination)	1 064	1 150	37.9	780	6.7	0.93	28	1.36	158
ν_μCC 0π (proton or other track as pion)	305	8 926	2.8	1 117	26.9	0.03	110	0.27	11
ν_μCC 1$\pi^\pm$ outside the measured phase space	52.1	675	3.8	384	17.1	0.08	14	0.14	3.05
ν_μCC with a π^0	77.8	442	31.0	3 756	2.7	0.18	2.51	0.02	29
ν_μCC $\geq 2\pi^\pm$, no π^0	178	231	77.3	885	4.2	0.77	2.30	0.20	42
out of fiducial volume	57.1	507	5.1	302	17.9	0.11	11	0.19	3.19
NC and ν_eCC	113	592	13.0	857	9.3	0.19	8.75	0.13	12
total ν-MC	1 848	12 523	171	8 080	84.9				
CC+NC with a π^0 (adopted π^0 target; overlaps two rows above)	83.3	488	33.1	4 118	3.6	0.17	2.52	0.020	23

The neutrino-simulation prediction of each region carries flux, cross-section, re-interaction and detector uncertainties of 21–35%, evaluated from the multisim weights and the Run-4 detector-variation samples on the files carrying the region flags. Its correlation with the signal-region yield across the flux universes is 0.98–1.00, so the flux term cancels in a transfer. Table 27 gives the uncertainty of the transfer factor of each target class, recomputed as a ratio in every universe and variation.

Table 27: Transfer factor T of each control region's target class and its fractional uncertainty (%), with T recomputed in every universe so that the numerator–denominator covariance is included. The flux term cancels to 2–3%; the cross-section term does not, because it changes the composition of the two regions differently; the detector term is the largest. For the adopted CC+NC π^0 class the Monte-Carlo-statistical floor of the detector entry is given in parentheses and replaces the entry in the total where it is larger.

Beam	Region	Target class	T	flux	cross section	re-int.	MC stat	detector	total
FHC	CC0π	ν_μCC 0π	0.0341	3.4	10.2	5.6	2.4	20.4	23.9
FHC	multi-π	ν_μCC $\geq 2\pi^\pm$	2.3600	3.1	4.5	2.8	4.8	14.7	16.6
FHC	π^0	ν_μCC with π^0	0.0195	2.6	6.0	1.7	4.3	16.1	18.0
FHC	CC0π	all neutrino MC	0.1412	1.7	12.8	2.5	0.9	9.2	16.1
FHC	π^0	CC+NC with π^0 (adopted)	0.0200	2.2	5.7	1.8	4.1	15.9 (11.1)	17.6
RHC	CC0π	ν_μCC 0π	0.0342	3.0	10.1	5.4	2.0	25.4	28.1
RHC	multi-π	ν_μCC $\geq 2\pi^\pm$	2.3043	1.8	3.7	2.7	3.8	19.8	20.7
RHC	π^0	ν_μCC with π^0	0.0207	1.8	6.5	1.8	3.2	11.8	14.1
RHC	CC0π	all neutrino MC	0.1476	2.1	12.2	2.4	0.8	9.7	15.9
RHC	π^0	CC+NC with π^0 (adopted)	0.0202	1.7	6.4	1.5	3.1	11.9 (20.3)	21.6

The detector entries are quadrature sums of eight differences between finite samples, so part of each is the statistics of the variation samples. Evaluating every shift on two disjoint halves of the generated events estimates that floor: it is 1–2% for the region yields, but for the transfer factors only the CC0π term (20–25% against a 6–7% floor) is a genuine detector sensitivity; the π^0 term is comparable to its floor (11–20%), and the multi-π term lies below it. A fitted correction therefore carries the larger of the transfer uncertainty and its floor.

Two limits of the control-region set. About 30% of the FHC neutrino background has no region: NC and ν_eCC without a π^0, out-of-fiducial-volume events and CC1$\pi^\pm$ outside the measured phase space, the last subtracted with a reconstruction bias of +30–66% for below-threshold pions (Table 8 of the analysis note). And the CC0π region constrains the CC0π rate, not the proton-as-pion misidentification that carries it into the signal region ($T = 0.03$, detector-dominated); an inverted pion-PID region would test that without touching the signal region and has not been defined.

20.4 The π^0 region with and without the final multiplicity cut

The frozen π^0 definition drops the final multiplicity cut (Sec. 19). A reprocess of the simulation, dirt and beam-off samples with the alternative flag, beam-on files untouched, compares the two. The cut removes 11% (FHC) and 14% (RHC) of the region and every component in nearly the same proportion, so the target purity changes by less than half a percent, the overlap with multi-π is unchanged and the transfer-factor uncertainties are unchanged within their precision (T itself rises by 12–16%). The leading-shower-energy shape changes by at most 0.7% of unit area in any bin. The frozen definition is kept as the larger sample. Its 11–14% of events that fail the signal-region multiplicity cut cannot populate the signal region, so a discrepancy confined to them would be attributed to the target class; the stored multiplicities allow item 4(i) of the procedure to locate it.

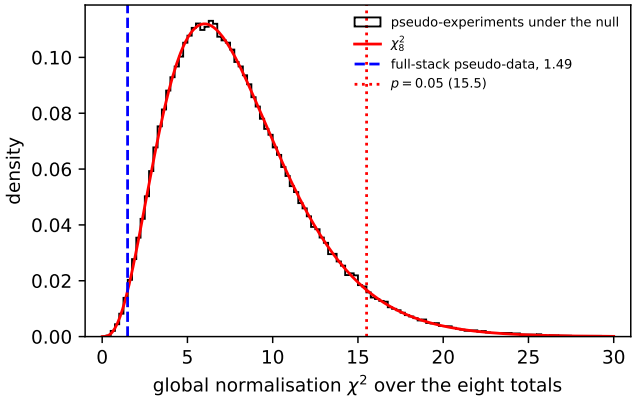


Figure 8: Null calibration of the global normalisation χ^2 over the eight (region, mode) totals: 2×10^5 pseudo-experiments with the prediction drawn from its pre-fit covariance and Poisson-fluctuated data, against the χ^2_8 density. The full-stack pseudo-data value, 1.49, is low because those pseudo-data carry none of the prediction's systematic variation.

20.5 Frozen procedure for the comparison with beam-on data

The procedure below is fixed before beam-on data in the control regions are examined.

1. **What is examined.** For each region the event total and the reconstructed p_μ, $\cos\theta_\mu$ and muon–pion opening angle, and for the π^0 region the leading shower energy, at the binning of the released observables where one exists and otherwise at the frozen binning (Fig. 7). FHC and RHC are examined separately, the combined configuration as a cross-check, and the totals also per run period. Data are compared with the complete prediction, neutrino simulation + beam-off + dirt.

2. **Statistics.** For each region and beam mode, against the pre-fit prediction: (a) the normalisation pull, with the summed covariance of the complete prediction, reported together with the pull against the statistical and detector terms alone, whose resolution (9–24%) is set by the detector term; (b) a shape χ^2 of the pre-declared distributions with the normalisation marginalised, so that the common-mode flux and cross-section terms do not make the test powerless (Sec. 14.5 of the analysis note); (c) a global normalisation χ^2 over the eight (region, mode) totals with the full pre-fit covariance, the single pass/fail statistic. On the full-stack pseudo-data the global χ^2 is 1.49/8, the sixteen pulls lie within ± 0.7 and all twenty-two shape tests have $p > 0.05$. A discrepancy confined to one region fails the set at 95% confidence beyond 12–13% of the CC0π or π^0 total and beyond 45–52% of the multi-π or cosmic total; a common shift of all eight totals, which the correlated flux, cross-section and detector terms absorb, fails it only beyond 50%. The regions test the background model region by region, not the overall normalisation, which the measurement carries as its flux and cross-section uncertainty.

3. **Acceptable agreement.** The set passes if the global χ^2 has $p > 0.05$. Under the null (2×10^5 pseudo-experiments, Fig. 8) the test fails at $p < 0.05$ in 5.1% and at $p < 0.01$ in 1.1% of cases; the sampling noise of the covariance is not re-thrown. A region with $|\text{pull}| > 2$ or shape $p < 0.05$ is flagged for item 4 even when the set passes; with sixteen pulls and twenty-two shape tests about two false flags are expected. A single-bin excursion beyond 3σ with a passing total is examined for a reconstruction pathology but does not flag the region.

4. **Responses.** Entered when the global test fails or a region is flagged; a flagged region within a passing set receives at most response (b). Regions are treated in the order CC0π, π^0,

then the two diagnostic regions, since the CC0π class is 15% of the π^0 region. (i) *Localised reconstruction failure.* A run period or a single bin beyond 3σ while the rest passes triggers a reconstruction-quality investigation, not a model correction. The period is excluded only if an independent detector or processing defect is established, which removes it from the comparison and from the measurement and makes a new frozen release; otherwise it stays in. No cut or binning is changed and no single bin is removed. (ii) *Fitted normalisation correction (c).* Only for CC0π and π^0 (target purity above 50% of the neutrino simulation) and only if the region's shape test passes: a one-parameter fit of the target class to FHC and RHC together, applied if it improves the fit at $p < 0.01$ in a likelihood-ratio test. The π^0 region meets the purity rule narrowly (50.1/51.0%), with a class and denominator settled after the purities were known, so a π^0 correction is considered only after the pre-fit result is documented. The two fitted regions are disjoint. The fit rescales the target class in every systematic universe, so that its correlation with the signal-region prediction is kept; its uncertainty is the transfer-factor uncertainty or its floor, whichever is larger, added in quadrature to the leverage of the components held fixed (the non-target background, beam-off, signal contamination and dirt, with shares 0.77, 0.16, 0.19, 0.03 of the target in the π^0 region and 0.27, 0.14, 0.13, 0.05 in CC0π). (iii) *Added uncertainty (b).* Otherwise the prediction is left unchanged and the full observed discrepancy, as a fraction of the target class, is assigned as a normalisation uncertainty on that class in the signal region through its transfer factor. (iv) *Model revision (d).* If the post-treatment global χ^2 still has $p < 0.05$, or a region remains flagged, the background model is revised before the signal region is opened. The pass/fail test is always the pre-fit one. FHC and RHC share one background model: a class that passes in one mode and fails in the other is treated under (b). The signal contamination of each region is held at the tune in any fit, which keeps the regions blind to the signal: a 30% signal excess would move the CC0π total by 2.7%, inside the 9–24% resolution of every pull. Multi-π and π^0 share events and are treated jointly.

5. **Prohibited responses.** Changing a selection cut or the binning of a released observable, choosing a generator or tune after inspecting the regions, bin-by-bin reweighting, and any use of the beam-on signal-region sample.

20.6 Scope: inclusive regions only

The procedure of Sec. 20.5 covers the four inclusive regions. The proton-tagged regions of §20.7 apply the proton tag rather than inverting it and therefore do not constrain the ≈13% proton mis-tag background that most affects the hadronic observables; the inverted-proton-PID region that would do so has not been implemented.

20.7 Control regions for the proton-tagged sample [PROTON-TAGGED]

The proton-tagged files carry the four inclusive region flags (Sec. 10.1 of the analysis note), so the regions can be evaluated with a tagged proton required, in the phase space of the measurement (Table 28). They are more signal-depleted than for the inclusive signal (5.6–7.2% contamination in CC0π and π^0). The tag costs about a third of the statistics but suppresses the beam-off contribution by about a factor six (FHC CC0π: 1 075 → 179), since cosmic-induced events rarely contain a reconstructed proton above threshold. The pseudo-data to simulation ratios of the two large regions stay at 0.98–1.01, so the per-run bookkeeping holds. The multi-π regions scatter within one standard deviation; the cosmic region of this throw lies 4.3 (FHC) and 3.7 (combined) standard deviations high, from Runs 1 and 5, while forty ensemble throws reproduce its expectation there. None of these regions constrains the mis-tag background, and cutting on a measured observable ($W_{\pi p}$, p_n, δp_T) cannot replace an inverted-PID region: the signal definition is inclusive in those variables, so such a cut splits the signal region (purity 47–53% in every slice) rather than depleting it.

Table 28: Control-region yields of the proton-tagged sample with a tagged proton required, full exposure. "Sig." is the proton-tagged signal contamination of the neutrino simulation; $N_{\text{data}}/N_{\nu\text{MC}}$ compares the pseudo-data with the neutrino simulation.

Beam	Region	N_{data}	Signal	ν bkg	beam-off	Sig. frac.	$N_{\text{data}}/N_{\nu\text{MC}}$
FHC	CC0π	7 250	409	6 896	179	5.6%	0.992
FHC	multi-π	48	9.6	45	0.2	17.7%	0.887
FHC	π^0	3 497	259	3 321	148	7.2%	0.977
FHC	cosmic	44	1.4	22	0.1	6.2%	1.895
RHC	CC0π	8 569	476	8 098	196	5.6%	0.999
RHC	multi-π	72	15	62	1.3	19.9%	0.935
RHC	π^0	4 902	333	4 534	162	6.8%	1.007
RHC	cosmic	39	2.4	30	0.5	7.4%	1.206
Comb	CC0π	15 819	885	14 994	375	5.6%	0.996
Comb	multi-π	120	25	106	1.5	19.0%	0.915
Comb	π^0	8 399	592	7 855	310	7.0%	0.994
Comb	cosmic	83	3.8	52	0.6	6.9%	1.494

20.8 Validation of the background fit

The background fit, the normalisation fit of the control-region target classes, is validated on simulation. Sets of 300 toys give pull means of -0.05 to 0.05, widths of 0.98–1.09 and coverage of 64–70%. On Asimov data, injected shifts of a class are recovered exactly, a 30% signal excess moves the normalisations by at most 0.13σ and a detector variation by at most 0.15σ. A likelihood-ratio test of common against beam-specific normalisations detects a +15% shift of the RHC simulation alone ($\Delta\chi^2 = 13.3/2$ on the totals). An adaptive Metropolis sampler reproduces the MIGRAD/MINOS intervals within 0.02 in the mean and 3% in the width. Ensembles of 10^5 toys show that with the combined Neyman–Pearson data variance the pull mean of the two-dimensional fits does not converge to zero (+0.02 to +0.03, six to ten standard errors), because that variance depends on the fluctuating data; the offset vanishes when the prediction is the data variance, and the fit uses that Pearson form.

21 Selection studies [INCLUSIVE]

21.1 Selection-optimisation study

Choice of phase space and opening-angle cut. The measured phase space and the $\theta_{\mu\pi} < 2.6$ rad cut were chosen on the Run-1 FHC sample, starting from a reference phase space of $p_\mu, p_\pi > 0.1\,\text{GeV}/c$. The numbers below and in Table 29 come from that study. The full-exposure yields, efficiency and purity of the released measurement are those of Table 11 of the analysis note.

- Signal MC (POT-weighted): 324.7 events
- Total background: 243.6 ($\approx$ 199.7 ν-bkg + 40.1 EXT + 3.8 dirt)
- True signal generated (efficiency denominator): 2577–2684 events (observable-dependent)
- Signal purity: 57.1%
- Signal efficiency: 12.1–12.6%
- Measured phase space (reported result): efficiency 15.6%, purity 54.9% (reference phase-space study: selected signal 311.9, generated 1999, background 256.4; Table 29)

Per-run scaling. The yields above and in Table 11 of the analysis note use the per-run POT factors of Table 5 of the analysis note (each run scaled by its own $D_{\text{run}}/\text{MC}_{\text{run}}$), and the EXT by the per-run trigger ratios (per-run trigger ratios, Sec. 5 of the analysis note). A single per-mode factor would differ from the per-run scheme by about 2% in the FHC final-cut signal, a small

and nearly uniform re-weighting of the run mixture that moves efficiency and purity by $< 0.5\%$. The per-run scheme is used because real beam-on data require it. The pseudo-data are thrown from the same per-run-scaled prediction, so the closure tests the bookkeeping of the scheme, not its correctness against data.

Measured phase space and efficiency turn-on. The reference selection has almost no efficiency at low momentum and at large opening angle. In the fine truth distributions the per-bin efficiency is only ∼2–5% for $p_\mu, p_\pi < 0.15\,\mathrm{GeV}/c$ (the 10/20 cm track-length cuts), and the region above the reco $\theta_{\mu\pi} = 2.6$ rad cut is dominated by cosmics (∼7% purity). The measured phase space $p_\mu > 0.15$, $p_\pi > 0.175\,\mathrm{GeV}/c$, $\theta_{\mu\pi} < 2.6$ rad (Sec. 6 of the analysis note) raises and flattens the efficiency to ∼13–20% (Fig. 20 of the analysis note). Table 29 compares the selection across the iterations, and it fixes the two opening-angle choices.

(i) The 2.6 rad upper cut is kept. On the full exposure the events it removes are exactly those of the cosmic control region (Table 13 of the analysis note): in FHC, 5.4 signal, 62 neutrino background, 42 beam-off and 19.2 dirt, a signal purity of 4.2%. Dropping the cut would lower the signal-region purity from 57.3% to 53.1%, raise its beam-off content from 24 to 66 events and recover only 2.7% of the generated signal (2055 → 1999 in the table). The removed region serves instead as the beam-off control region.

(ii) No lower opening-angle cut is used. The small-angle region has high purity (∼58%), so a lower cut reduces the selected signal without improving purity.

Table 29: Selection-optimisation study across the phase-space iterations, on the Run-1 FHC sample. Every entry is Monte-Carlo derived (GENIE overlay signal and background, EXT, dirt); the thresholds chosen here are those of the full-exposure selection of Table 11 of the analysis note. The beam-off column uses the pooled beam-off scaling of that study, so its absolute values are not those of the full exposure; the ranking of the options is unaffected. "no upper cut" drops the $\theta_{\mu\pi} < 2.6$ cut; "[0.314, 2.6]" adds a lower OA cut. The final choice keeps the 2.6 rad upper cut with no lower cut.

	Reference (p>0.1)	no upper cut (p>.15/.175)	[0.314, 2.6] (lower cut)	**Final** (θ<2.6)
Generated signal	2684	2055	1916	1999
Selected signal	324.7	320.3	301.8	311.9
Efficiency	12.1%	15.6%	15.8%	**15.6%**
EXT (cosmic)	40.1	136.2	38.0	40.1
Total bkg	243.6	385.4	252.3	256.4
Purity	**57.1%**	45.4%	54.5%	**54.9%**
sig × purity	185.6	145.4	164.4	**171.2**

21.2 Momentum-threshold validation

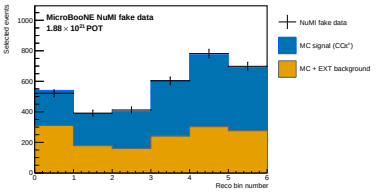
The two signal-definition momentum thresholds sit at the selection efficiency turn-on, measured against a threshold-free true signal (the CC ν_μ $1\mu1\pi^\pm$ topology with the momentum/angle cuts removed):

Threshold	below	above	turn-on
$p_\mu > 0.15$ GeV/c	∼0–1.5%	∼9–12%	0.145→0.155
$p_\pi > 0.175$ GeV/c	∼0–3%	∼12–16%	0.165→0.185

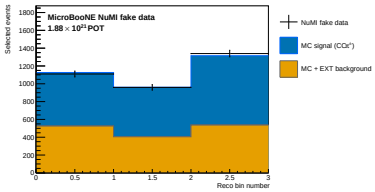
Both are correctly placed: lowering them would add near-zero-efficiency phase space, raising them would discard measurable signal. The reconstructed pion momentum (muon-hypothesis range) is biased low and increasingly so with momentum (mean reco−true ≈ 0 at threshold, -0.15 GeV/c at 0.40) from the μ-vs-π mass hypothesis plus pion re-interaction; a pion-hypothesis mass correction is applied to the p_π reco bins.

21.3 Combined-configuration reconstruction-level spectra

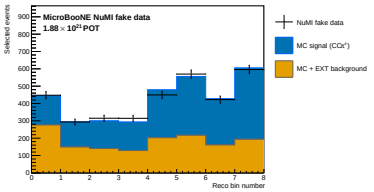
The two figures below show the selected and background-subtracted spectra of the combined configuration; the FHC and RHC spectra are in the analysis note.



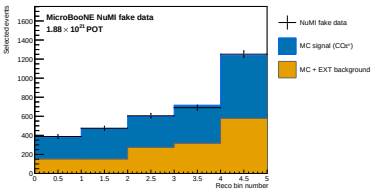
(a) p_μ



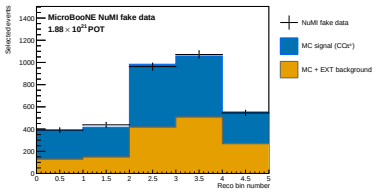
(b) p_π



(c) $\cos\theta_\mu$

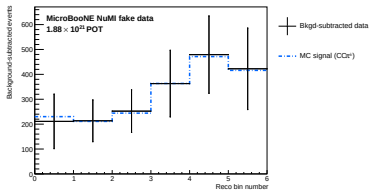


(d) $\cos\theta_\pi$

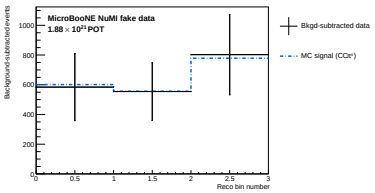


(e) $\theta_{\mu\pi}$

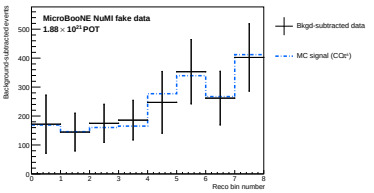
Figure 9: Stage 1, combined: selected reconstructed spectra in the combined configuration.



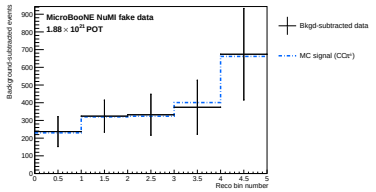
(a) p_μ



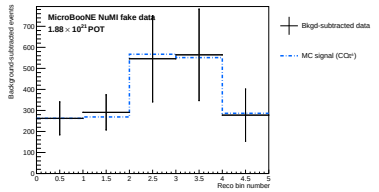
(b) p_π



(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$



(e) $\theta_{\mu\pi}$

Figure 10: Stage 2, combined: background-subtracted data compared with the central-value MC signal.

22 Analysis pipeline and automated guards [SHARED]

Every released number is produced by the XSecAnalyzer chain in four stages that are re-runnable from the raw PeLEE ntuples:

1. the event processing applies the selection(s) to each PeLEE ntuple and writes the processed event files, for the neutrino overlays, the dirt overlay and the per-run beam-off samples;
2. the universe builder constructs, per observable and configuration, the true and reconstructed histograms of every systematic universe from the files listed in that configuration;
3. the unfolder sums the universes with the per-run exposure scaling, builds the covariance, unfolds, and writes the closure sidecar, the systematics dump and the matrix tables;
4. the export step writes the data release (curves, A_C matrices, covariance decompositions); the figure macros draw from the closure sidecars; the one-bin total uses the same chain with the single-bin configurations; the count-level tables (cut flow, cut-and-count check, control-region yields) are produced with the same weights and scalings.

Five automated guards protect the normalisation, the currency of released artefacts, the result tables and the blinding:

- A normalisation manifest reads the authoritative configuration files and asserts that exposure, flux, target count and beam-off scaling agree across all configurations, and that every declared systematic is backed by a file.
- A coupling assertion in the unfolder refuses a configuration whose declared detector variations are not present in its file list.
- A staleness check refuses to ship any released artefact or result figure older than the extraction it depicts, and compares the integrals printed on the p_π figures with their sidecars.
- A table check parses every result table of the analysis notes and this supplement and fails if a value disagrees with the released results table, if a table does not sit under its own label, or if a row's column count differs from its header.
- A blinding guard treats a beam-on file without MC truth as real data and refuses it unless an explicit unblinding switch is set; the same detection selects the beam-off treatment (added to pseudo-data, subtracted from real data), so no manual switch exists to forget.

The release was built from the processed ntuples in a single pass with every cached universe deleted, so no intermediate result is carried over. A clean-cache rebuild of the same kind reproduced a previous release bin by bin to 10^{-9} (recorded in the change log); that comparison has not been repeated on the current release.

Part III
Rejected alternatives and development history

The alternatives below were studied and not adopted; none is used in the results. The change log gives the chronology.

23 The first binning optimisation [INCLUSIVE]

The first optimisation required bins at least as wide as the local detector resolution and holding at least 45 selected signal events. It gave p_μ seven bins, p_π, $\cos\theta_\mu$ and $\cos\theta_\pi$ five and $\theta_{\mu\pi}$ seven, and lowered the total uncertainty of every observable (p_μ from 33 to 25%, $\cos\theta_\pi$ from 49 to 34%), chiefly through the statistical and unfolding-amplified terms. Where A_C left no resolved information, $\cos\theta_\pi$ and $\theta_{\mu\pi}$ were then coarsened to four and five bins. A re-check against the 0.68 criterion, with pre-screened candidates extracted in full, found that p_μ, $\cos\theta_\mu$, θ_μ and p_π could not go finer, that further $\theta_{\mu\pi}$ bins carried no resolved information (the smallest singular value of A_C went to zero), and that $\cos\theta_\pi$ admitted a fifth bin in the forward hemisphere, which improved the conditioning of A_C from 0.27 to 0.46 (FHC) and was adopted. The binning review of §7, with edges optimised at every bin count on the corrected exposure, revised these conclusions and replaced the binnings of this section (§2).

24 The muon angle: $\cos\theta_\mu$ versus θ_μ [INCLUSIVE]

The muon angle is reported only as $\cos\theta_\mu$. The same partition of the angle retains up to one degree of freedom more as $\cos\theta_\mu$ than as θ_μ, because the regulariser acts on the density (§7), so a θ_μ result adds no information. In the five-bin binnings that preceded the review, A_C suppressed the integral of the forward-peaked $\cos\theta_\mu$ strongly (row sums 0.17–0.39 in FHC, against 0.66–0.91 for p_μ). The suppression follows the conditioning of the response, not the shape of the truth: across the observables the integral factor of A_C is uncorrelated with peakedness, and the flattest generators are smeared no differently from the others. With the width-aware operator the tune, from which the pseudo-data are thrown, fits θ_μ best ($\chi^2 = 2.14$ on five bins, against 5.55–9.10 for the generators); under the uniform stencil every model gave 2.8–4.7, because every model had been mapped onto the same curve (§11).

25 Extending the pion threshold below 0.175 GeV/c [INCLUSIVE]

A $[0.113, 0.175]$ GeV/c bin was added to the pion-momentum binning as a test, lowering the phase space to the ν_e threshold. Its efficiency is 2.2%, an order of magnitude below the 15–17% of the other bins, as the reconstruction turn-on implies. The extraction stayed well behaved (closure $\chi^2 = 1.41/6$, flux term unchanged) at the price of a large uncertainty in the new bin, and GiBUU could not be re-binned because its generator-level event record was not retained. The threshold remains $p_\pi > 0.175$ GeV/c.

26 The rejected five-bin p_π response [INCLUSIVE]

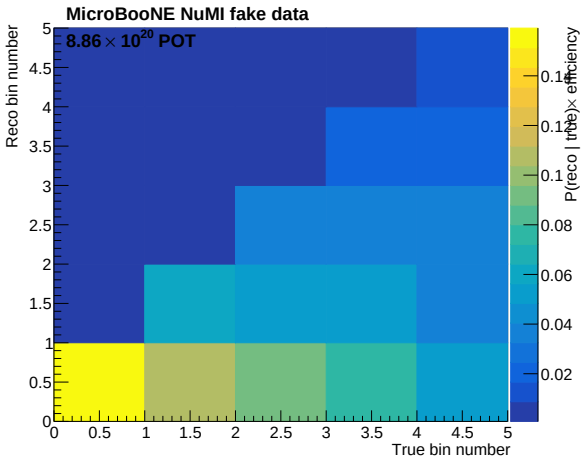


Figure 11: Response matrix of the rejected five-bin p_π scheme (FHC, analysis sample): reconstructed events pile into the lowest bin from every true bin (§6.2), which motivated the merged regions of §6.2.5.

27 Golden-pion classifier and other pion-momentum estimators [INCLUSIVE]

All four inputs of the BNB golden-pion classifier are available here, the track wiggliness as `trk_avg_deflection_stdev_v`. Golden pions are the wigglier ones, by a factor of two in the mean, because a pion that stops scatters strongly in its last centimetres while one that interacts is cut short while still fast; a cut at 0.001 keeps 87% of golden pions and raises the golden fraction from 40 to 53%. In the trained classifier (§6.2.6) the deflection variables dominate and the Bragg ratios contribute almost nothing. Tight working points appear better than perfect truth selection because a hard cut keeps long tracks and so cuts on the reconstructed momentum itself; they are not used. Golden pions still reconstruct at 0.73 of the true momentum, a momentum-dependent bias that no global rescaling repairs.

The other estimators reach the same limit. The MCS momentum is unbiased on average in the top bin but has an RMS of 210% there, because these contained, interacting pions are 25–84 cm long and MCS needs of order a metre; as a classifier input the range–MCS difference ranks high but leaves the ROC integral unchanged, the signature of a redundant input. Calorimetry is truncated at the interaction point like range, with three to five times its RMS. Above 0.8 GeV/c golden and non-golden pions reconstruct identically, since a pion that stops inside the TPC at that momentum has already lost most of its energy. A BDTG regression removes most of the bias ($W_{\pi p}$ bias from -0.127 to -0.005 GeV/c^2) but recovers only a quarter of the attainable resolution (0.119 against a ceiling of 0.046) and moves migration from the top bins into the first; trained on simulation and upstream of every observable, it is not adopted. Adding the energy of the pion daughters recovers of order 10% of the missing energy at high momentum, because absorption on argon puts much of it into neutrons and into tracks too short to reconstruct; it raises the best upper diagonal only from 5.5 to 15.7% and is not adopted.

28 Cosmic-direction cuts [INCLUSIVE]

Beam-off events are enriched at backward reconstructed muon angles (32% at $\cos\theta_\mu < 0$ against 4% of the signal), but a cut does not pay. The beam-off background, 24 of the 1 522 selected FHC

events, is subtracted without bias, so removing all of it would reduce the statistical uncertainty of the subtraction by about 4 events; any cut that costs more than about 0.5% of the signal is a net loss. A $\cos\theta_\mu < 0$ cut would remove about 7 beam-off and 33 signal events. The calorimetric direction flag of the CC0π analysis is bounded by the same ceiling.

References

[1] X.-G. Lu *et al.* (MINERvA Collaboration), "Measurement of final-state correlations in neutrino muon-proton mesonless production on hydrocarbon at $\langle E_\nu \rangle = 3$ GeV," Phys. Rev. Lett. **121**, 022504 (2018).